

# **Selected Applications in Natural Language Processing, Image Analysis, Board Games, and Edge Computing**

Ole-Christoffer Granmo  
Centre for AI Research  
University of Agder  
Norway

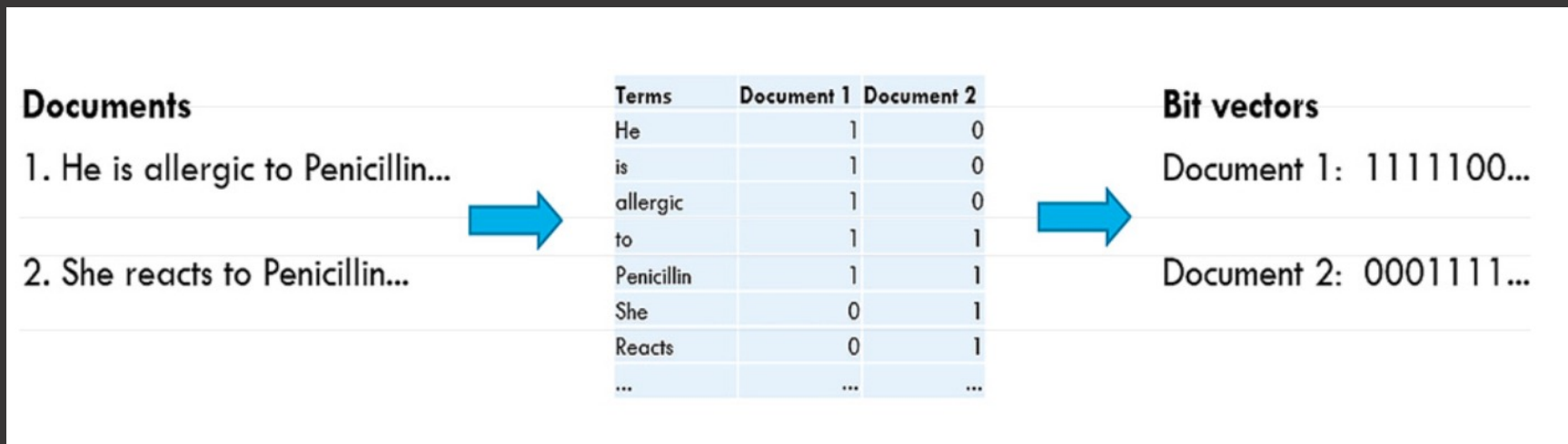
# Overview

- Natural language processing
- Image analysis and convolution
- Board games
- Edge AI



# Natural language processing

# Set of words

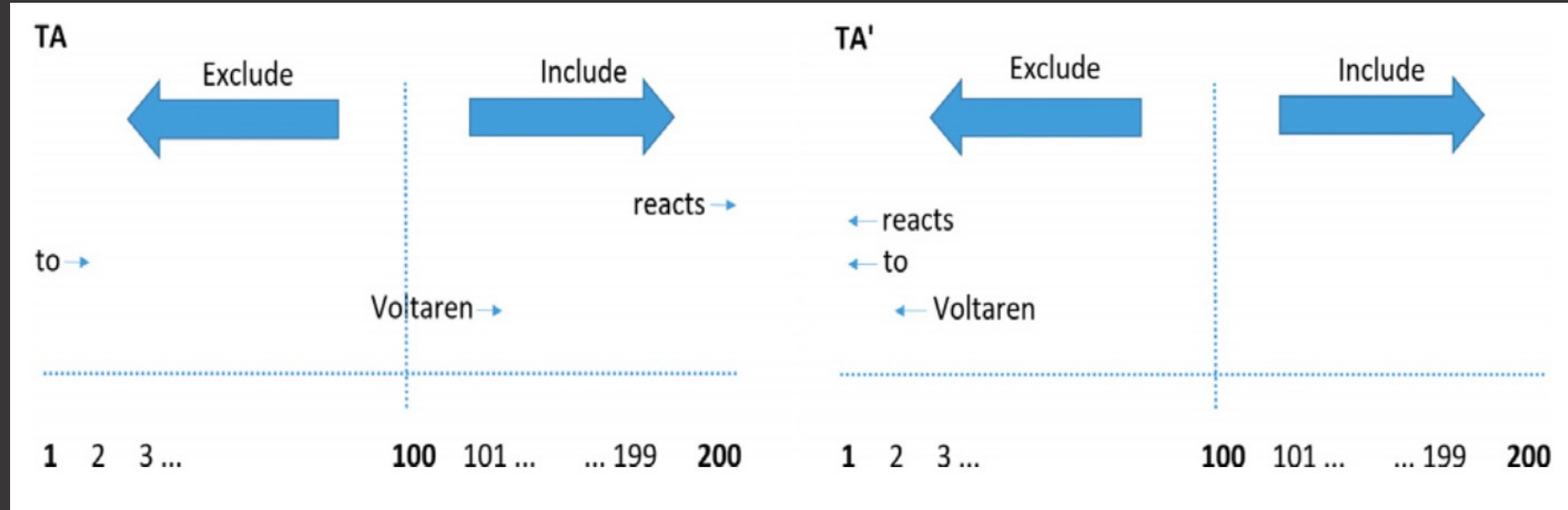




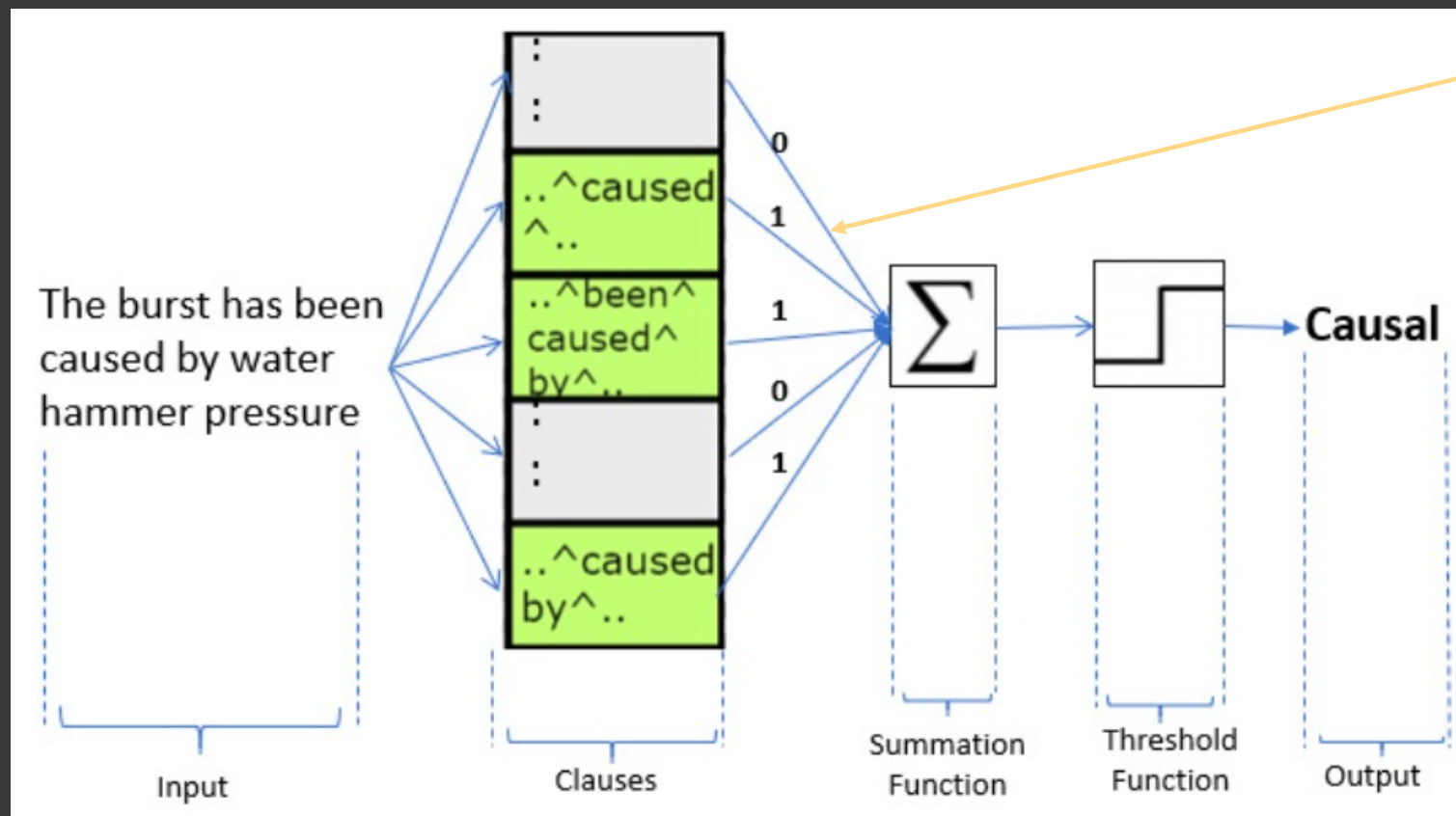
# Linguistic patterns

*ending*  $\wedge$  *boring*  $\wedge$   $\neg$ *good*  $\wedge$   $\neg$ *interesting*  $\wedge$   $\neg$ *like*,  $\dots$

# Learning natural language concepts



# Recognizing concepts with rules



Local interpretability



Saha, R., Granmo, O.C., Goodwin, M. (2023). Using Tsetlin Machine to discover interpretable rules in natural language processing applications. Expert Systems, 40(4)

Global interpretability

$\text{Sentence\_Contains}((\text{caused\_by}) \vee (\text{causes}) \vee (\text{triggered\_by}) \vee (\text{resulted\_in}) \vee (\text{been\_caused\_by}) \vee (\text{radiating\_from}) \vee (\text{lead\_to}) \vee \dots) \Rightarrow \text{Class}(\text{Cause-Effect}).$



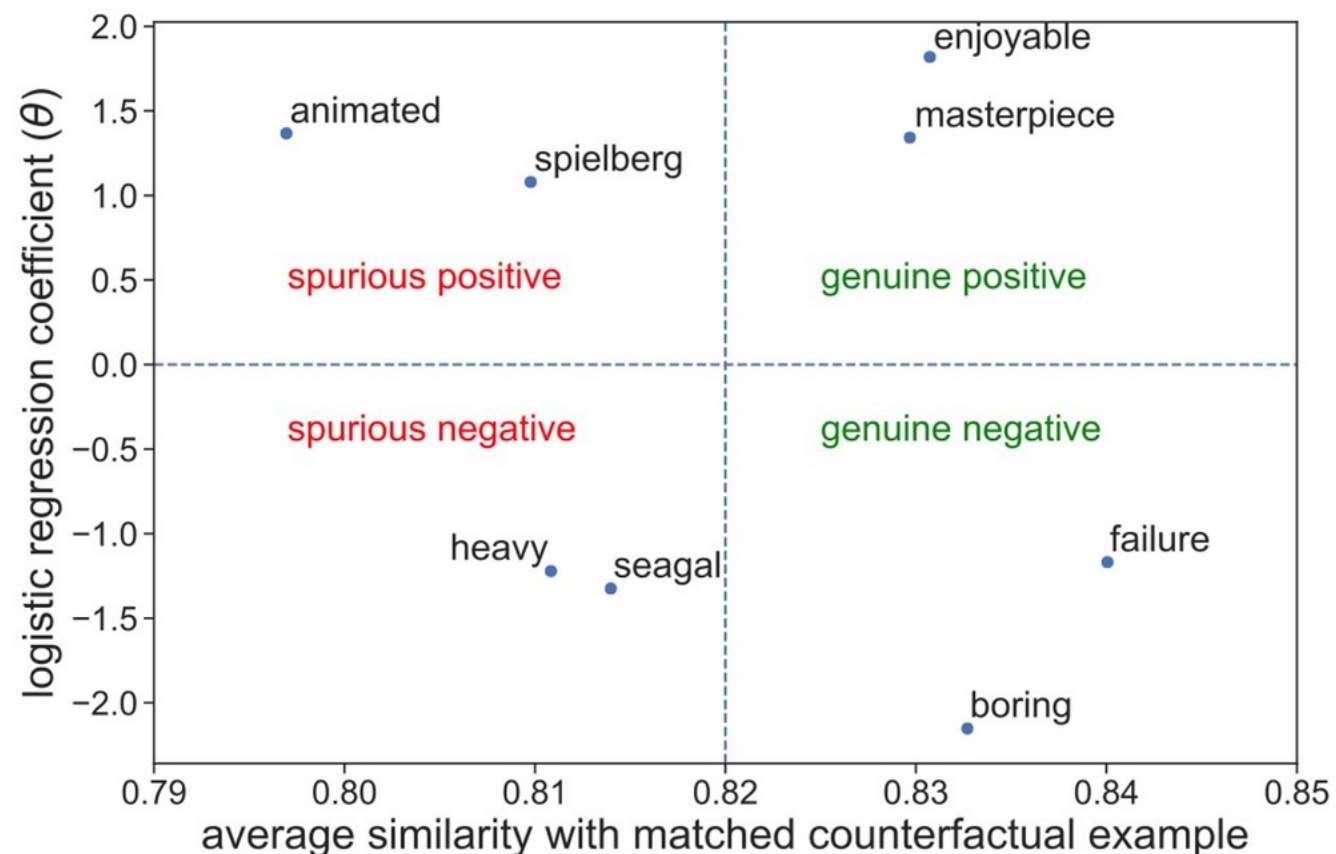
Saha, R., Myhre S. (2022). Interpretable Text Classification in Legal Contract Documents Using Tsetlin Machines. *First International Symposium on the Tsetlin Machine*

TABLE III  
TEXTUAL CONTEXT AND RELEVANT CLAUSE LITERALS FROM SOME EXAMPLE CONTRACTS FOR THE CLASS “EXPIRATION DATE”. TERMS IN BOLD CORRESPOND TO LITERALS THAT CONSTITUTE CLAUSES THAT CLASSIFIED THE TEXT CORRECTLY

| Doc. Id | Contract Name                                                                                   | Relevant Text                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Literals of Relevant Clauses                                                                                             |
|---------|-------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------|
| 199     | BUFFALOWILDWINGSINC_06_05_1998-EX-10.3-FRANCHISE AGREEMENT                                      | The <b>term of</b> this Agreement is for ten (10) years commencing on the date of <b>this Agreement</b> , <b>unless terminated</b> as provided by <b>this Agreement</b> .                                                                                                                                                                                                                                                                                                                                                                                  | 1. term_of AND this_Agreement<br>AND unless_terminated                                                                   |
| 194     | VAPOTHERM, INC. - Manufacturing and Supply Agreement                                            | The <b>term of this Agreement</b> is three years from and including the date of <b>this Agreement</b> (the “Initial Term”), with automatic renewal for additional successive one-year terms (each a “ <b>Renewal Term</b> ” and together with the Initial Term, the “Term”) <b>unless</b> no later than [* * *] days prior to the end of the Initial Term, or any <b>Renewal Term</b> either party notifies the other that it wishes to <b>terminate this Agreement</b> effective the end of the Initial Term or that <b>Renewal Term</b> , as applicable. | 2. term_of AND this_Agreement<br>AND unless AND terminate<br>3. Initial_Term AND terminate<br>4. Renewal_Term AND unless |
| 198     | HYPERIONSOFTWARECORP_09_28_1994-EX-10.47-EXCLUSIVE DISTRIBUTOR AGREEMENT                        | This Agreement shall have an initial term of one (1) year from the date first above written (the “Initial Term”), and shall thereafter automatically renew for successive two (2) year periods (each a “ <b>Renewal Term</b> ”), <b>unless earlier terminated</b> in accordance with the terms of this Agreement.                                                                                                                                                                                                                                          | 4. Renewal_Term AND unless<br>5. unless_earlier AND<br>earlier_terminated                                                |
| 191     | EuromediaHoldingsCorp_20070215_10SB12G_EX-10.B(01)_525118_EX-10.B(01)_Content License Agreement | The term of this Agreement (the “ <b>Initial Term</b> ”) shall commence as of the Effective Date and, <b>unless earlier terminated</b> in accordance with this Agreement, shall <b>terminate</b> on June 30, 2010.                                                                                                                                                                                                                                                                                                                                         | 3. Initial_Term AND terminate<br>5. unless_earlier AND<br>earlier_terminated                                             |
| 211     | NeoformaInc_19991202_S-1A_EX-10.26_5224521_EX-10.26_Co-Branding Agreement                       | INITIAL TERM shall mean the Effective Date through the day prior to the second anniversary of the Effective Date, <b>unless earlier terminated</b> pursuant to Section 11.                                                                                                                                                                                                                                                                                                                                                                                 | 5. unless_earlier AND<br>earlier_terminated                                                                              |
| 217     | LeadersonlineInc_20000427_S-1A_EX-10.8_4991089_EX-10.8_Co-Branding Agreement                    | Term shall mean the <b>Effective Date through</b> June 15 , 2001 and any Renewal Term ( as defined in paragraph 7.4 herein . )                                                                                                                                                                                                                                                                                                                                                                                                                             | 6. Effective_Date<br>AND through                                                                                         |
| 245     | ExactSciencesCorp_20180822_8-K_EX-10.1_11331629_EX-10.1_Promotion Agreement                     | This Agreement shall be effective as of the <b>Effective Date</b> and shall continue in effect <b>through</b> December 31 , 2021 and any <b>Renewal Term</b> ( the “Term” ) , <b>unless terminated</b> earlier as set forth herein.                                                                                                                                                                                                                                                                                                                        | 6. Effective_Date AND through<br>7. Renewal_Term AND<br>unless_terminated                                                |

| Categories                  | TM    |       |       |       | CNN-BiLSTM |       |       |       | FastText |       |       |       | BERT-Base |       |       |       | RoBERTa |       |       |       |
|-----------------------------|-------|-------|-------|-------|------------|-------|-------|-------|----------|-------|-------|-------|-----------|-------|-------|-------|---------|-------|-------|-------|
|                             | P     | R     | F1    | Acc   | P          | R     | F1    | Acc   | P        | R     | F1    | Acc   | P         | R     | F1    | Acc   | P       | R     | F1    | Acc   |
| Anti Assignment             | 95.57 | 94.05 | 94.65 | 94.05 | 0.23       | 4.85  | 0.45  | 4.85  | 90.09    | 74.64 | 81.35 | 74.64 | 97.02     | 94.5  | 95.39 | 94.5  | 97.78   | 96.32 | 96.82 | 96.32 |
| Audit Rights                | 94.04 | 95.04 | 94.27 | 95.04 | 90.31      | 90.1  | 90.21 | 90.1  | 89.58    | 55.27 | 66.99 | 55.27 | 95.83     | 95.43 | 95.6  | 95.43 | 96.21   | 94.76 | 95.28 | 94.76 |
| Cap on Liability            | 94.36 | 94.57 | 94.46 | 94.57 | 0.19       | 4.38  | 0.37  | 4.38  | 89.9     | 73.52 | 80.56 | 73.52 | 96.55     | 95.08 | 95.68 | 95.08 | 97.73   | 97.89 | 97.78 | 97.89 |
| Effective Date              | 96.27 | 96.79 | 96.47 | 96.79 | 96.79      | 94.76 | 95.61 | 94.76 | 94.16    | 60.79 | 72.92 | 60.79 | 97.35     | 95.19 | 96.06 | 95.19 | 98.13   | 94.39 | 95.8  | 94.39 |
| Exclusivity                 | 90.21 | 94.83 | 92.47 | 94.83 | 85.3       | 81.16 | 83.13 | 81.16 | 90.52    | 75.23 | 81.8  | 75.23 | 90.9      | 92.44 | 90.56 | 92.44 | 93.86   | 94.09 | 93.96 | 94.09 |
| Expiration Date             | 95.51 | 93.87 | 94.57 | 93.87 | 0.15       | 3.88  | 0.29  | 3.88  | 92.4     | 62.42 | 73.63 | 62.42 | 97.65     | 94.16 | 95.38 | 94.16 | 98.24   | 96.96 | 97.39 | 96.96 |
| Insurance                   | 96.23 | 97.08 | 95.84 | 97.08 | 93.58      | 92.0  | 92.72 | 92.0  | 93.46    | 70.79 | 80.41 | 70.79 | 99.32     | 99.2  | 99.23 | 99.2  | 98.98   | 98.9  | 98.93 | 98.9  |
| License Grant               | 94.98 | 95.15 | 95.06 | 95.15 | 93.87      | 28.94 | 38.28 | 28.94 | 88.91    | 71.12 | 78.26 | 71.12 | 97.09     | 96.06 | 96.41 | 96.06 | 97.56   | 95.47 | 96.14 | 95.47 |
| Renewal Term                | 94.88 | 97.4  | 96.12 | 97.4  | 89.84      | 93.01 | 91.4  | 93.01 | 95.08    | 68.45 | 79.11 | 68.45 | 98.7      | 98.66 | 98.68 | 98.66 | 99.77   | 99.76 | 99.76 | 99.76 |
| Revenue Profit Sharing      | 93.78 | 95.62 | 93.82 | 95.62 | 86.17      | 87.01 | 86.57 | 87.01 | 92.29    | 62.16 | 73.16 | 62.16 | 93.87     | 91.99 | 92.7  | 91.99 | 97.63   | 97.69 | 97.65 | 97.69 |
| Termination for Convenience | 94.53 | 97.23 | 95.86 | 97.23 | 93.15      | 93.15 | 93.15 | 93.15 | 94.69    | 81.57 | 87.45 | 81.57 | 97.21     | 97.04 | 97.12 | 97.04 | 98.16   | 97.78 | 97.93 | 97.78 |
| Volume Restriction          | 95.77 | 97.86 | 96.81 | 97.86 | 86.74      | 92.31 | 89.44 | 92.31 | 95.14    | 60.33 | 73.57 | 60.33 | 85.73     | 91.18 | 88.37 | 91.18 | 93.48   | 94.51 | 93.91 | 94.51 |

# Spurious correlations



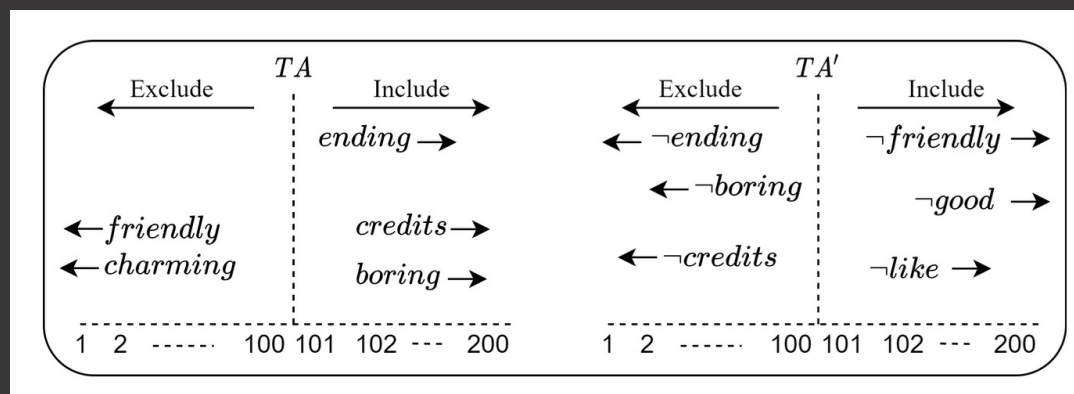
Wang, Zhao and Aron Culotta. "Identifying spurious correlations for robust text classification." *Findings* (2020).



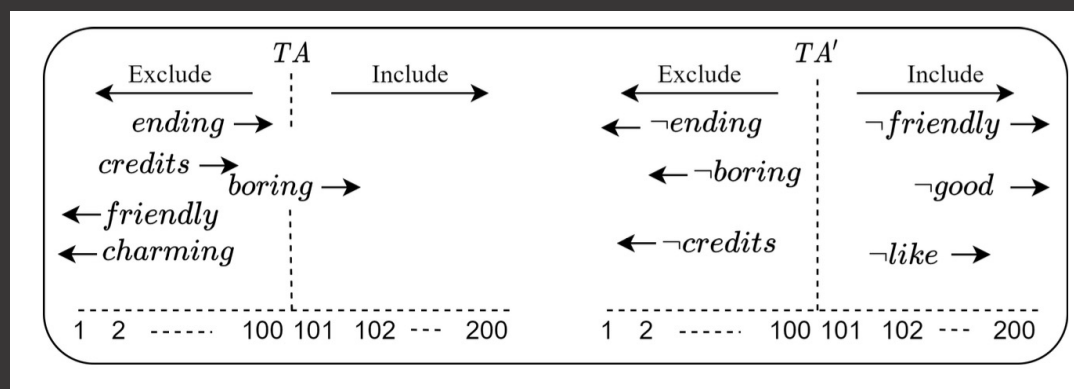
# Reasoning by elimination using negation

$\neg \text{friendly} \wedge \neg \text{charming} \wedge \neg \text{fascinating}, \dots$

Large  $s$



Small  $s$



Rohan Kumar Yadav, Lei Jiao, Ole-Christoffer Granmo, Morten Goodwin: Robust Interpretable Text Classification against Spurious Correlations Using AND-rules with Negation. IJCAI 2022.

# Robust reasoning by elimination using negation

|                                                                                       |                                                                                               |           |           |
|---------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------|-----------|-----------|
| $S_1$ Long, boring, blasphemous. Never have I been so glad to see ending credits roll | Negative Class Clauses                                                                        |           |           |
|                                                                                       | $C_1 = ending \wedge boring \wedge \neg good \wedge \neg interesting \wedge \neg like, \dots$ | $S_1$     | $S_2$     |
|                                                                                       | $C_3 = \neg fascinating \wedge \neg interesting \wedge \neg like, \dots$                      | $C_1 = 1$ | $C_1 = 0$ |
|                                                                                       | $C_5 = \neg good \wedge \neg friendly \wedge \neg like, \dots$                                | $C_3 = 1$ | $C_3 = 1$ |
|                                                                                       | $C_7 = \neg friendly \wedge \neg charming \wedge \neg fascinating, \dots$                     | $C_5 = 1$ | $C_5 = 0$ |
| $S_2$ How truly friendly, charming and cordial is this unpretentious old serial       | $C_9 = \neg excellent \wedge \neg good \wedge \neg like, \dots$                               | $C_7 = 1$ | $C_7 = 0$ |
|                                                                                       | Positive Class Clauses                                                                        |           |           |
|                                                                                       | $C_1 = truly \wedge \neg depressing \wedge \neg long \wedge \neg worst, \dots$                | $C_9 = 1$ | $C_9 = 1$ |
|                                                                                       | $C_3 = \neg bad \wedge \neg uninteresting \wedge \neg boring, \dots$                          | $S_2$     | $S_1$     |
|                                                                                       | $C_5 = \neg boring \wedge \neg worst \wedge \neg pretentious, \dots$                          | $C_1 = 1$ | $C_1 = 0$ |
|                                                                                       | $C_7 = \neg bad \wedge \neg charmeless \wedge \neg regret, \dots$                             | $C_3 = 1$ | $C_3 = 0$ |
|                                                                                       | $C_9 = serial \wedge \neg worse \wedge \neg unpleasant \wedge \neg like, \dots$               | $C_5 = 1$ | $C_5 = 0$ |
|                                                                                       |                                                                                               | $C_7 = 1$ | $C_7 = 1$ |
|                                                                                       |                                                                                               | $C_9 = 1$ | $C_9 = 0$ |

|                                                                                            |                                                                                               |            |            |
|--------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------|------------|------------|
| $S_1^{cf}$ Long, fascinating, soulful. Never have I been so sad to see ending credits roll | Negative Class Clauses                                                                        |            |            |
|                                                                                            | $C_1 = ending \wedge boring \wedge \neg good \wedge \neg interesting \wedge \neg like, \dots$ | $S_1^{cf}$ | $S_2^{cf}$ |
|                                                                                            | $C_3 = \neg fascinating \wedge \neg interesting \wedge \neg like, \dots$                      | $C_1 = 0$  | $C_1 = 0$  |
|                                                                                            | $C_5 = \neg good \wedge \neg friendly \wedge \neg like, \dots$                                | $C_3 = 0$  | $C_3 = 1$  |
|                                                                                            | $C_7 = \neg friendly \wedge \neg charming \wedge \neg fascinating, \dots$                     | $C_5 = 1$  | $C_5 = 1$  |
| $S_2^{cf}$ How truly depressing, charmless, and unpleasant is this pretentious old serial  | $C_9 = \neg excellent \wedge \neg good \wedge \neg like, \dots$                               | $C_7 = 0$  | $C_7 = 1$  |
|                                                                                            | Positive Class Clauses                                                                        |            |            |
|                                                                                            | $C_1 = truly \wedge \neg depressing \wedge \neg long \wedge \neg worst, \dots$                | $C_9 = 1$  | $C_9 = 1$  |
|                                                                                            | $C_3 = \neg bad \wedge \neg uninteresting \wedge \neg boring, \dots$                          | $S_2^{cf}$ | $S_1^{cf}$ |
|                                                                                            | $C_5 = \neg boring \wedge \neg worst \wedge \neg pretentious, \dots$                          | $C_1 = 0$  | $C_1 = 0$  |
|                                                                                            | $C_7 = \neg bad \wedge \neg charmeless \wedge \neg regret, \dots$                             | $C_3 = 1$  | $C_3 = 1$  |
|                                                                                            | $C_9 = serial \wedge \neg worse \wedge \neg unpleasant \wedge \neg like, \dots$               | $C_5 = 0$  | $C_5 = 1$  |
|                                                                                            |                                                                                               | $C_7 = 0$  | $C_7 = 1$  |
|                                                                                            |                                                                                               | $C_9 = 0$  | $C_9 = 0$  |



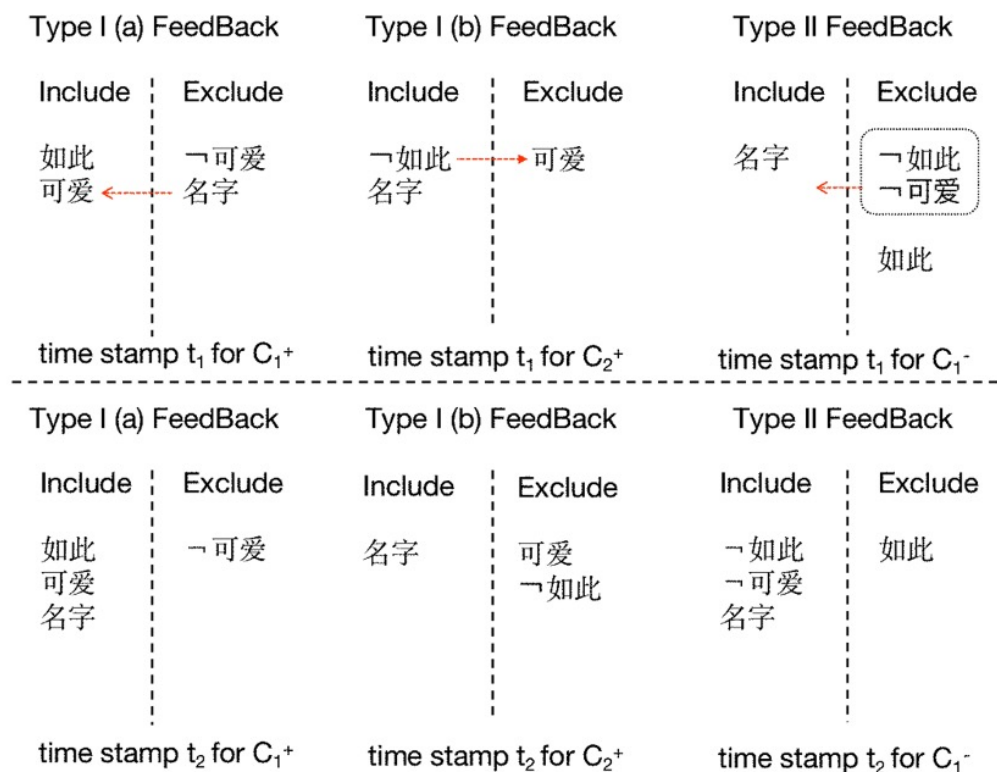
# Robust reasoning by elimination and negation

| Model                                                    | Word's Weightage in the Sentence                                                                                                                                                                                                                                                                                               | Sentiment Label                           |
|----------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------|
| Bi-LSTM with Attention Visualization                     | Really good movie. Maybe the best I've ever seen. Alien invasion, a la The Blob, with crazy good acting. Meteorite turns beautiful woman into a host body for nasty tongue. Engaging plot, great tongue. Absurd comedy worth watching. Maybe don't wash your hair or take out the trash but take time out to watch this movie. | Original: Positive<br>Predicted: Negative |
| Word's weightage based on clause score (TM with $s=20$ ) | Really good movie. Maybe the best I've ever seen. Alien invasion, a la The Blob, with crazy good acting. Meteorite turns beautiful woman into a host body for nasty tongue. Engaging plot, great tongue. Absurd comedy worth watching. Maybe don't wash your hair or take out the trash but take time out to watch this movie. | Original: Positive<br>Predicted: Negative |
| Word's weightage based on clause score (TM with $s=2$ )  | Really good movie. Maybe the best I've ever seen. Alien invasion, a la The Blob, with crazy good acting. Meteorite turns beautiful woman into a host body for nasty tongue. Engaging plot, great tongue. Absurd comedy worth watching. Maybe don't wash your hair or take out the trash but take time out to watch this movie. | Original: Positive<br>Predicted: Positive |

# Fake news detection in Chinese

|                        | Acc          | F1           |
|------------------------|--------------|--------------|
| SVM                    | 0.821        | 0.821        |
| LR                     | 0.823        | 0.814        |
| DPCNN [37]             | 0.819        | 0.802        |
| Bi-LSTM                | 0.823        | 0.818        |
| RCNN [38]              | 0.815        | 0.804        |
| BERT+Attention [36]    | 0.855        | 0.850        |
| <b>Tsetlin Machine</b> | <b>0.869</b> | <b>0.866</b> |

| Clause  | id | Chinese Text      | English Translation               |
|---------|----|-------------------|-----------------------------------|
| Clause+ | 0  | 全靠 ^ 粉丝 ^ 滤镜 ^ 扛着 | supported ^ by ^ fan ^ filter     |
|         | 1  | 全靠 ^ 粉丝 ^ 滤镜 ^ 扛着 | supported ^ by ^ fan ^ filter     |
|         | 2  | 全靠 ^ 粉丝 ^ 滤镜 ^ 扛着 | supported ^ by ^ fan ^ filter     |
| Clause- | 0  | 最棒 ^ 喜剧           | best ^ comedy                     |
|         | 1  | 恶心 ^ 片子 ^ 喜剧      | disgusting ^ movie ^ screenwriter |
|         | 2  | 尴尬 ^ 演技           | awkward ^ acting                  |



# Demo – reasoning by elimination

```
_LOGGER.info("Preparing dataset")
train, test = keras.datasets.imdb.load_data(num_words=args.imdb_num_words,
                                             index_from=args.imdb_index_from)

train_x, train_y = train
test_x, test_y = test

word_to_id = keras.datasets.imdb.get_word_index()
word_to_id = {k: (v + args.imdb_index_from) for k, v in word_to_id.items()}
word_to_id["<PAD>"] = 0
word_to_id["<START>"] = 1
word_to_id["<UNK>"] = 2
_LOGGER.info("Preparing dataset.... Done!")

_LOGGER.info("Producing bit representation...")

id_to_word = {value: key for key, value in word_to_id.items()}

training_documents = []
for i in range(train_y.shape[0]):
    terms = []
    for word_id in train_x[i]:
        terms.append(id_to_word[word_id].lower())

    training_documents.append(terms)

testing_documents = []
for i in range(test_y.shape[0]):
    terms = []
    for word_id in test_x[i]:
        terms.append(id_to_word[word_id].lower())

    testing_documents.append(terms)

vectorizer_X = CountVectorizer(
    tokenizer=lambda s: s,
    token_pattern=None,
    ngram_range=(1, args.max_ngram),
    lowercase=False,
    binary=True
)
```

```
X_train = vectorizer_X.fit_transform(training_documents)
feature_names = vectorizer_X.get_feature_names_out()
Y_train = train_y.astype(np.uint32)

X_test = vectorizer_X.transform(testing_documents)
Y_test = test_y.astype(np.uint32)
_LOGGER.info("Producing bit representation... Done!")

_LOGGER.info("Selecting Features....")

SKB = SelectKBest(chi2, k=args.features)
SKB.fit(X_train, Y_train)

selected_features = SKB.get_support(indices=True)
X_train = SKB.transform(X_train).toarray().astype(np.uint32)
X_test = SKB.transform(X_test).toarray().astype(np.uint32)

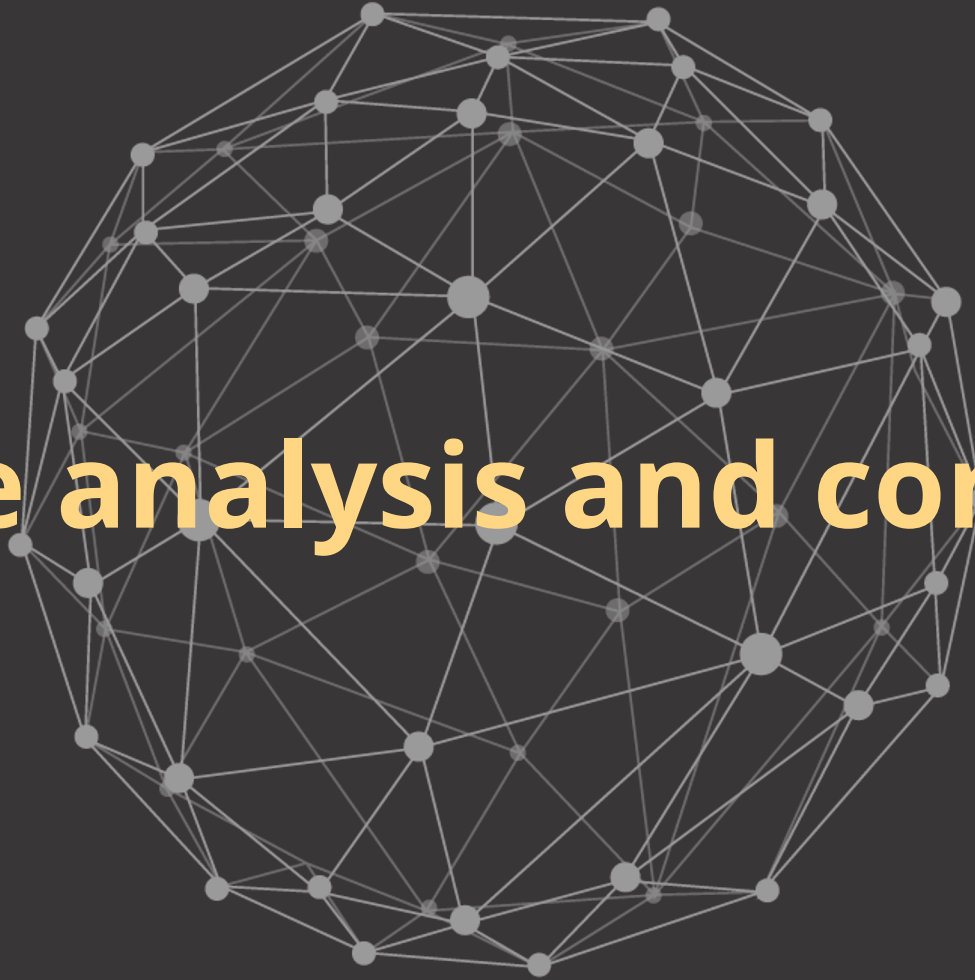
_LOGGER.info("Selecting Features.... Done!")

tm = TMClassifier(args.num_clauses, args.T, args.s, platform=args.device,
                 weighted_clauses=args.weighted_clauses, clause_drop_p=args.clause_drop_p)

_LOGGER.info(f"Running {TMClassifier} for {args.epochs}")
for epoch in range(args.epochs):
    benchmark1 = BenchmarkTimer(logger=_LOGGER, text="Training Time")
    with benchmark1:
        tm.fit(X_train, Y_train)

    benchmark2 = BenchmarkTimer(logger=_LOGGER, text="Testing Time")
    with benchmark2:
        result = 100 * (tm.predict(X_test) == Y_test).mean()

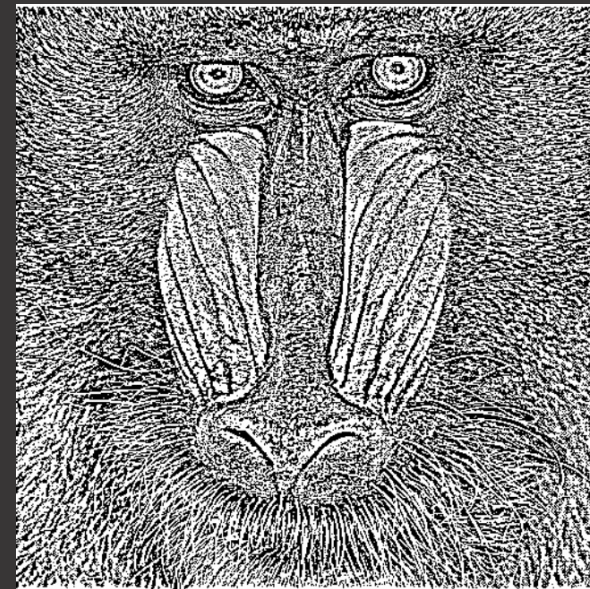
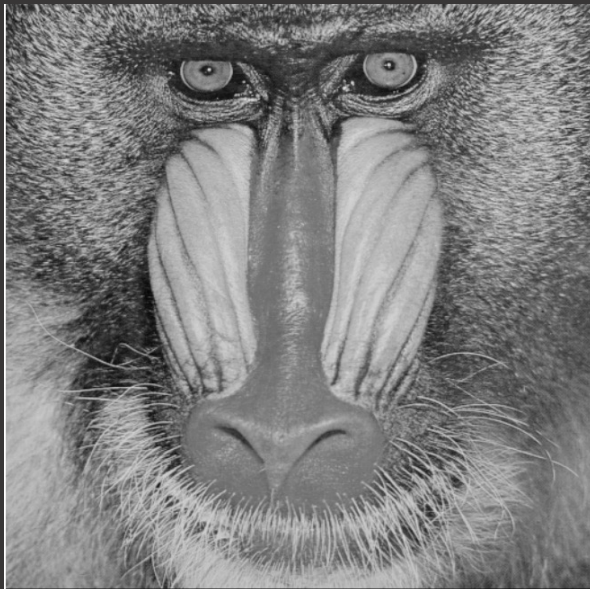
    _LOGGER.info(f"Epoch: {epoch + 1}, Accuracy: {result:.2f}, Training Time:
                 {benchmark1.elapsed():.2f}s, |f"Testing Time: {benchmark2.elapsed():.2f}s")
```



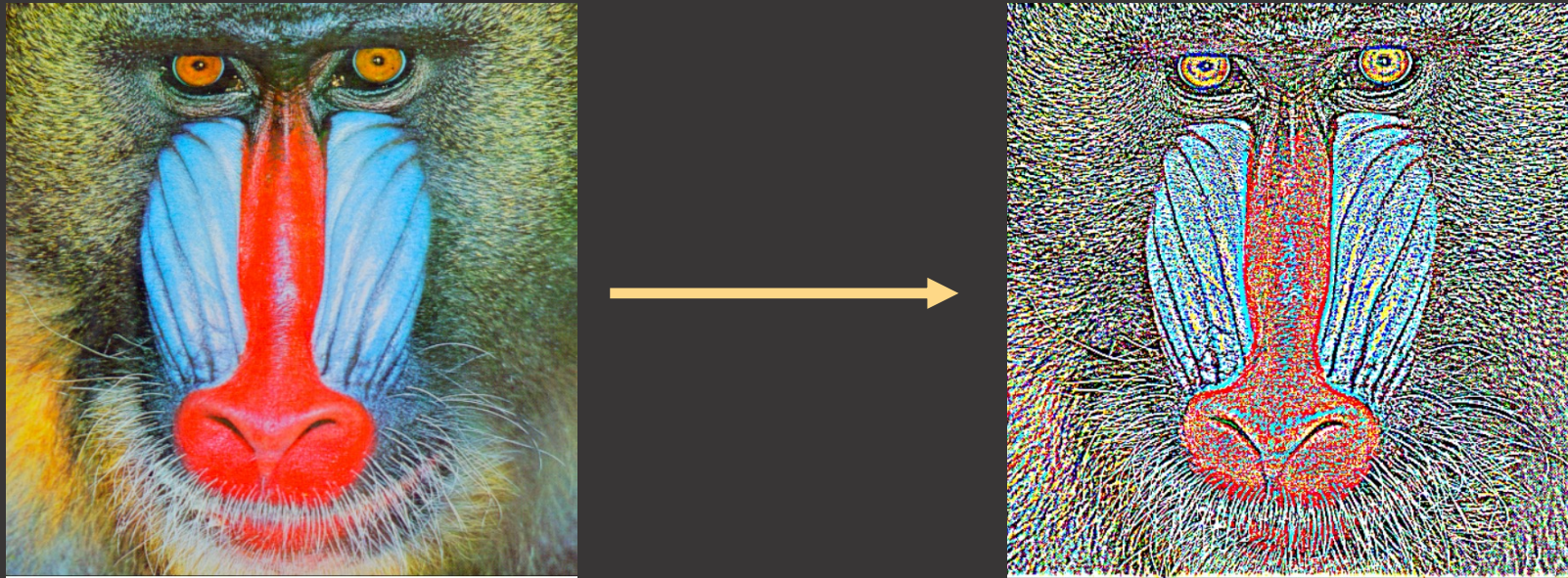
# Image analysis and convolution



# Booleanization with adaptive Gaussian thresholding



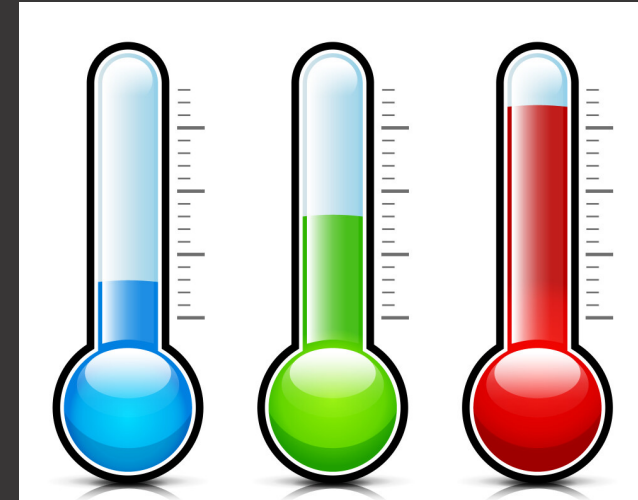
# Booleanization with adaptive Gaussian thresholding



```
for i in range(X_train.shape[0]):  
    for j in range(X_train.shape[3]):  
        X_train[i,:,:,:j] = cv2.adaptiveThreshold(X_train[i,:,:,:j], 1, cv2.ADAPTIVE_THRESH_GAUSSIAN_C, cv2.THRESH_BINARY, 11, 2)
```

# Booleanization with color thermometers

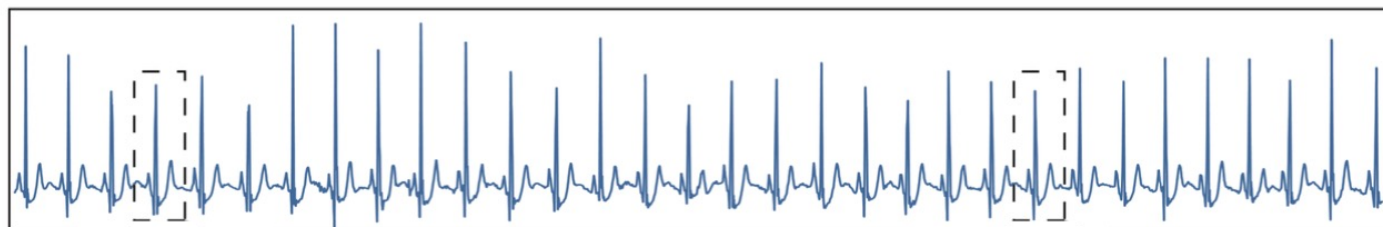
| Pixel Value, $V$   | Encoded value |
|--------------------|---------------|
| $V < 28$           | 00000000      |
| $28 \leq V < 56$   | 00000001      |
| $56 \leq V < 85$   | 00000011      |
| $85 \leq V < 113$  | 00000111      |
| $113 \leq V < 141$ | 00001111      |
| $141 \leq V < 170$ | 00011111      |
| $170 \leq V < 198$ | 00111111      |
| $198 \leq V < 226$ | 01111111      |
| $226 \leq V$       | 11111111      |



```
for z in range(resolution):
    threshold = (z + 1) * 255 / (resolution + 1)
    X_train[:, :, :, z] = X_train_org >= threshold
    X_test[:, :, :, z] = X_test_org >= threshold
```

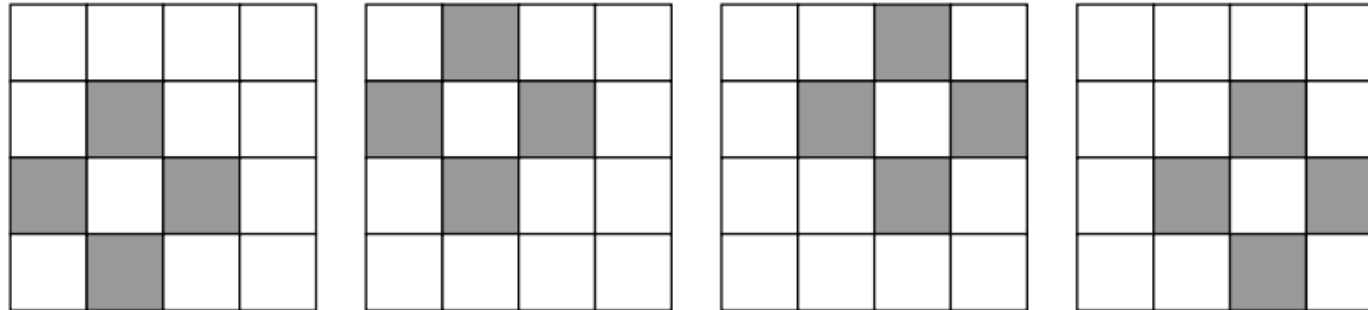


# Recognizing patterns in time and space

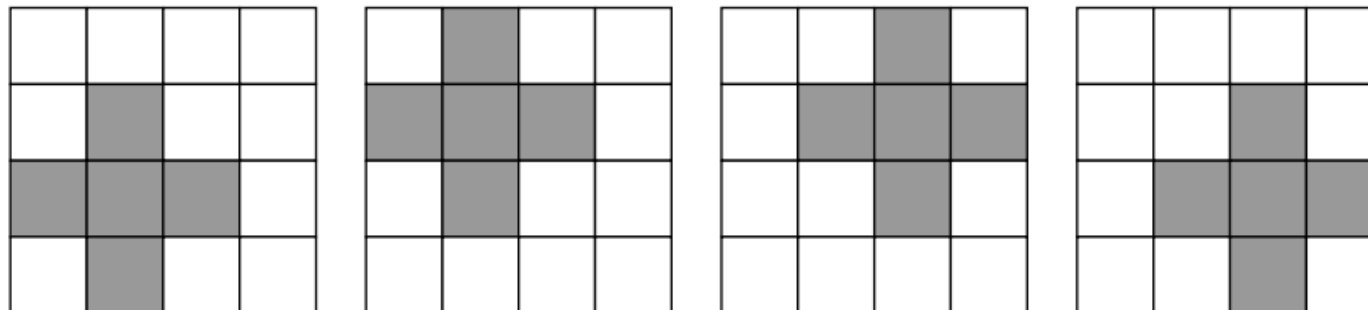


# Binary image input

Circle



Cross

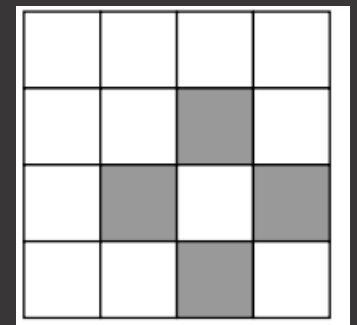
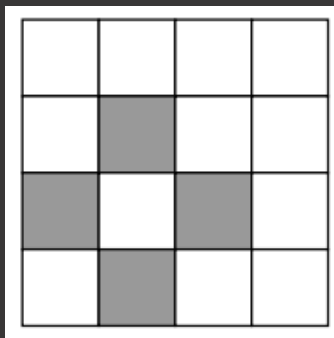
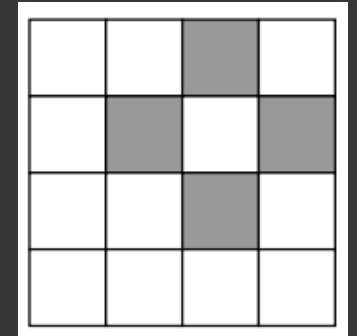
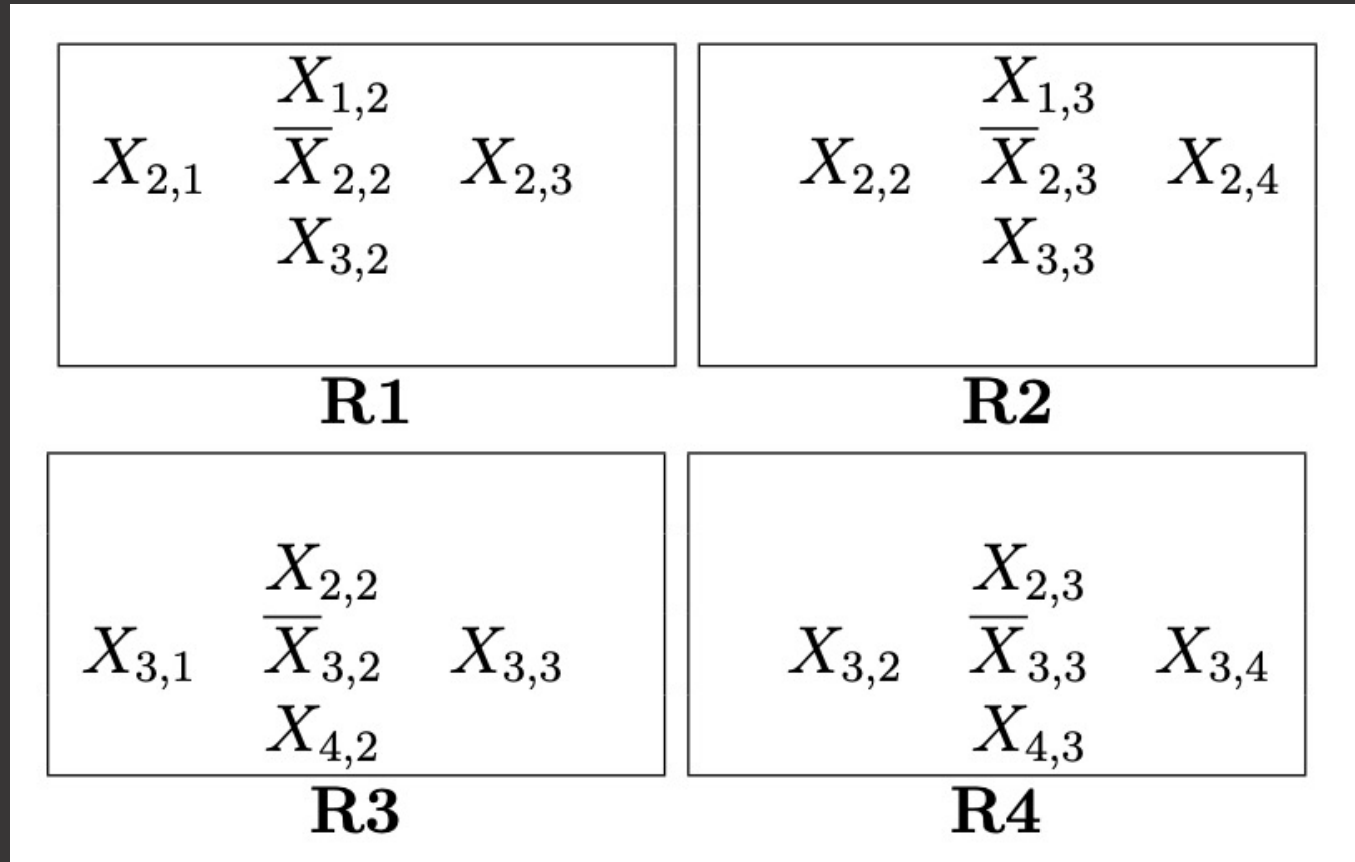
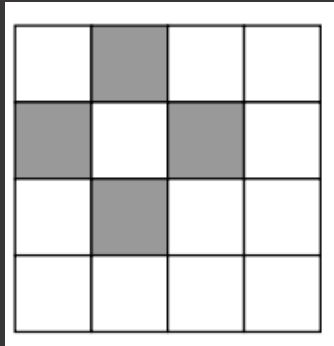


# Input and patterns

|           |           |           |           |
|-----------|-----------|-----------|-----------|
| $X_{1,1}$ | $X_{1,2}$ | $X_{1,3}$ | $X_{1,4}$ |
| $X_{2,1}$ | $X_{2,2}$ | $X_{2,3}$ | $X_{2,4}$ |
| $X_{3,1}$ | $X_{3,2}$ | $X_{3,3}$ | $X_{3,4}$ |
| $X_{4,1}$ | $X_{4,2}$ | $X_{4,3}$ | $X_{4,4}$ |

**R1:** if  $X_{1,2}$  and  $X_{2,1}$  and not  $X_{2,2}$  and  $X_{2,3}$  and  $X_{3,2}$  then *Circle*.

# Four rules recognizing circle in four locations



# Convolution with local window

|           |           |           |   |
|-----------|-----------|-----------|---|
| $X_{1,1}$ | $X_{1,2}$ | $X_{1,3}$ | · |
| $X_{2,1}$ | $X_{2,2}$ | $X_{2,3}$ | · |
| $X_{3,1}$ | $X_{3,2}$ | $X_{3,3}$ | · |
| ·         | ·         | ·         | · |

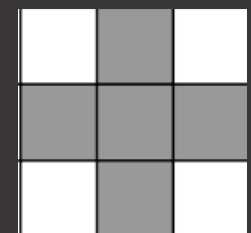
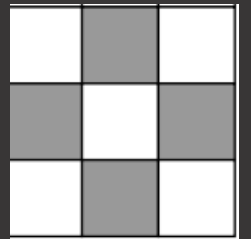
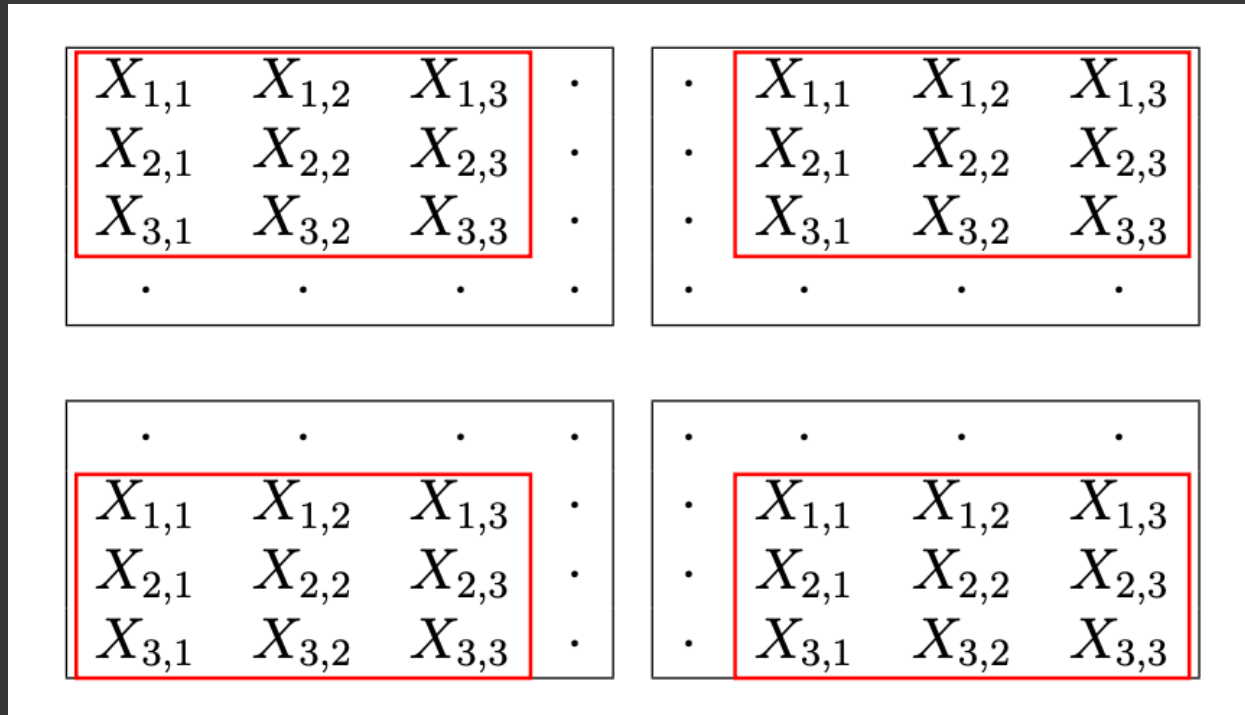
|   |           |           |           |
|---|-----------|-----------|-----------|
| · | $X_{1,1}$ | $X_{1,2}$ | $X_{1,3}$ |
| · | $X_{2,1}$ | $X_{2,2}$ | $X_{2,3}$ |
| · | $X_{3,1}$ | $X_{3,2}$ | $X_{3,3}$ |
| · | ·         | ·         | ·         |

|           |           |           |   |
|-----------|-----------|-----------|---|
| ·         | ·         | ·         | · |
| $X_{1,1}$ | $X_{1,2}$ | $X_{1,3}$ | · |
| $X_{2,1}$ | $X_{2,2}$ | $X_{2,3}$ | · |
| $X_{3,1}$ | $X_{3,2}$ | $X_{3,3}$ | · |

|   |           |           |           |
|---|-----------|-----------|-----------|
| · | ·         | ·         | ·         |
| · | $X_{1,1}$ | $X_{1,2}$ | $X_{1,3}$ |
| · | $X_{2,1}$ | $X_{2,2}$ | $X_{2,3}$ |
| · | $X_{3,1}$ | $X_{3,2}$ | $X_{3,3}$ |

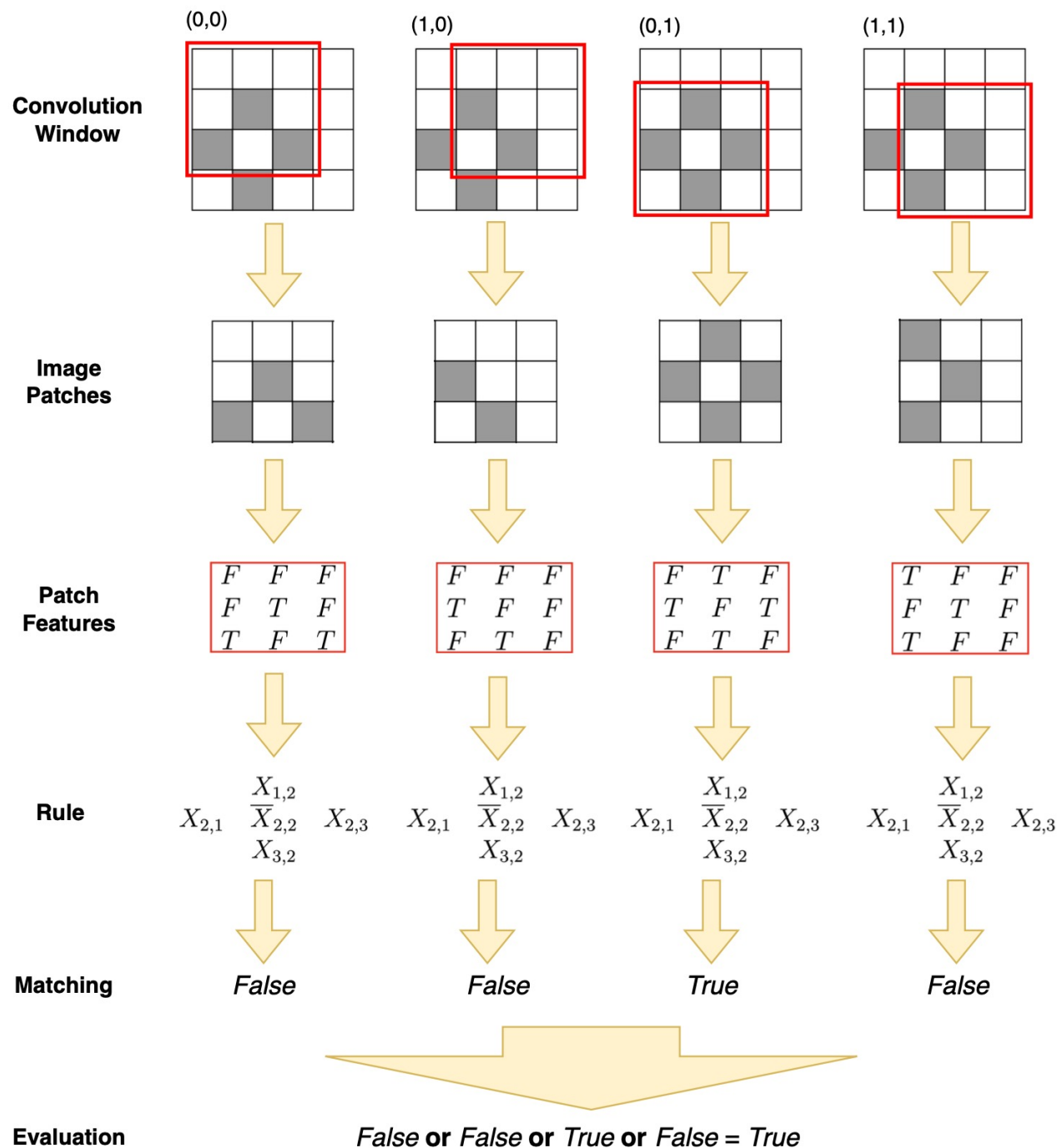


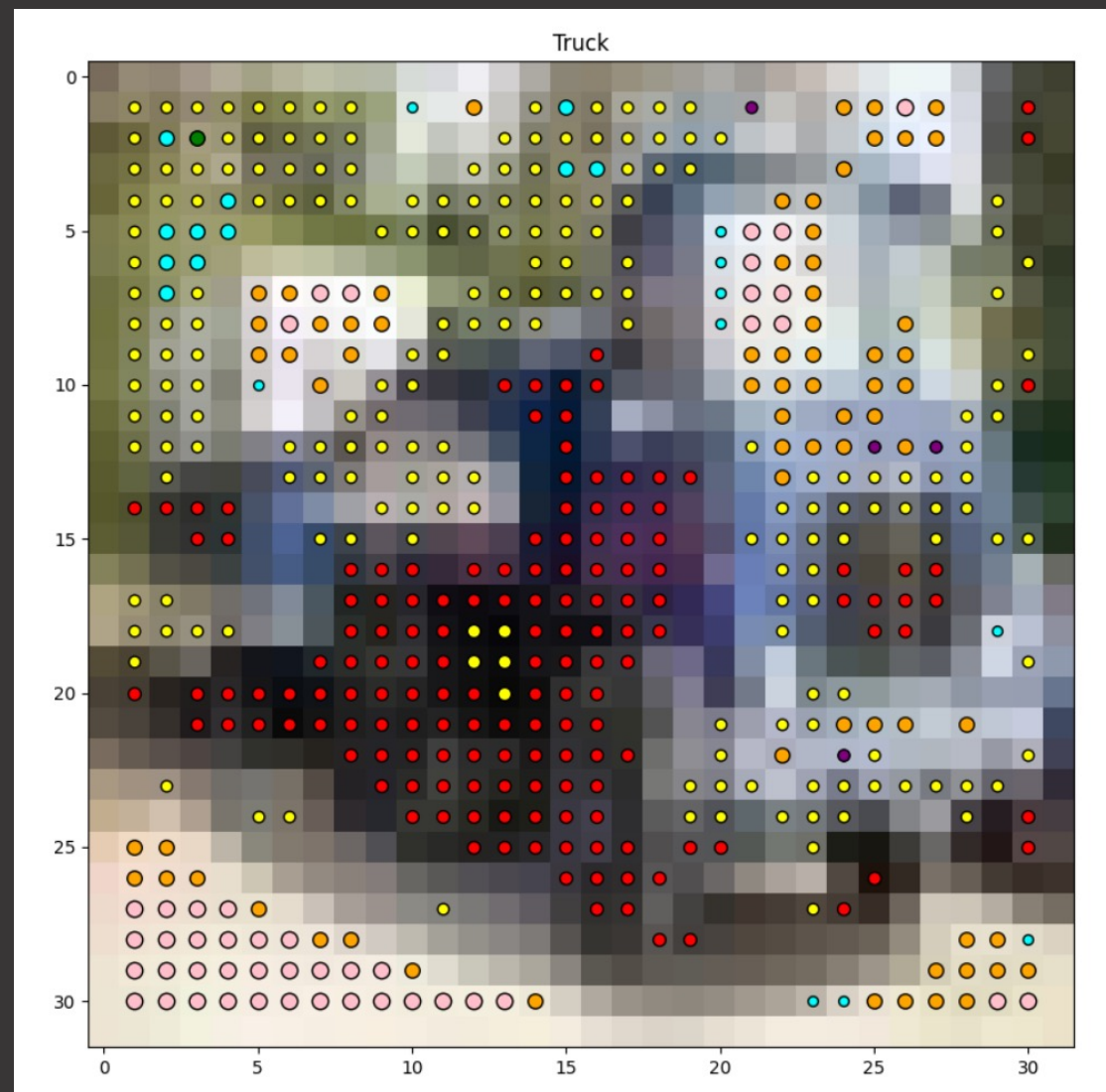
# Convolution with local window



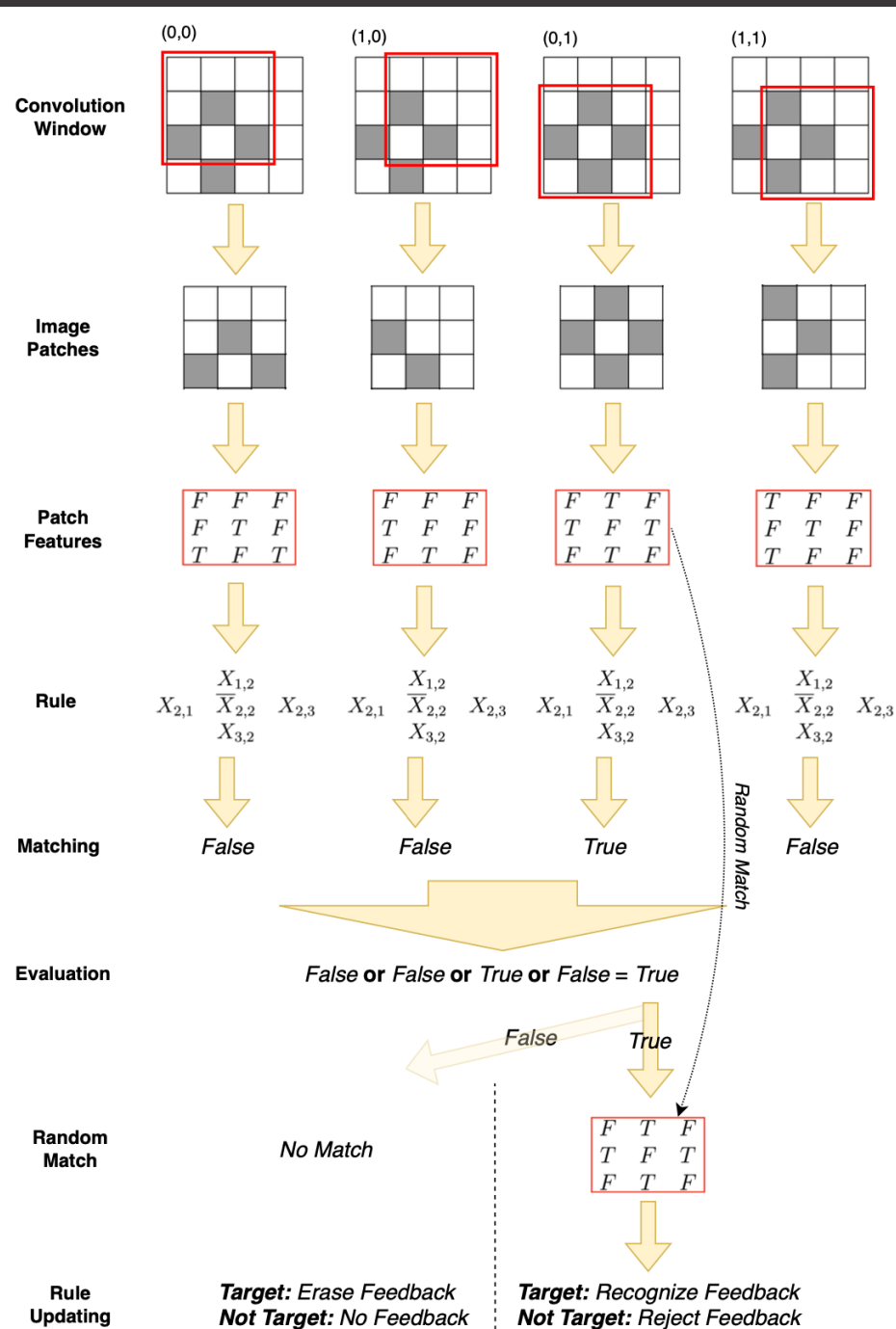
**R5:** if  $X_{1,2}$  and  $X_{2,1}$  and not  $X_{2,2}$  and  $X_{2,3}$  and  $X_{3,2}$  then *Circle*.

**R6:** if  $X_{1,2}$  and  $X_{2,1}$  and  $X_{2,2}$  and  $X_{2,3}$  and  $X_{3,2}$  then *Cross*.

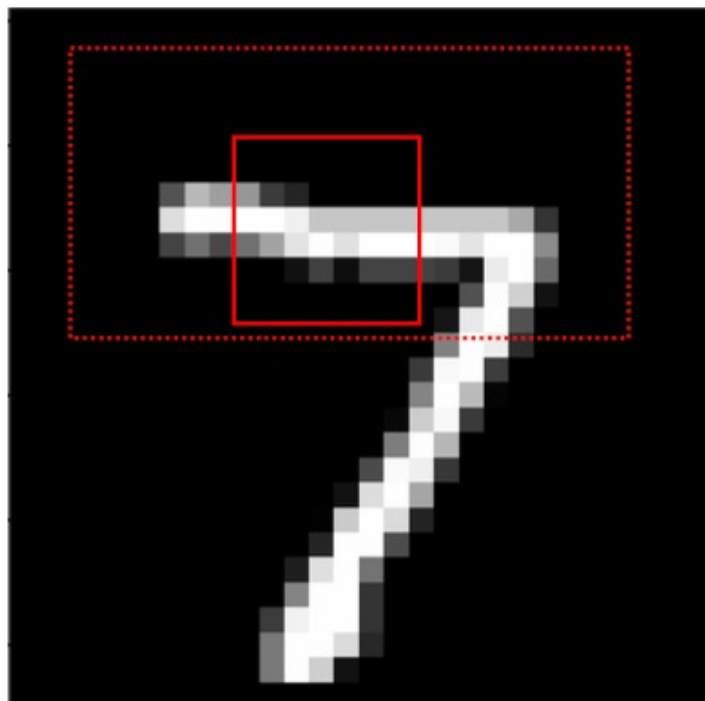




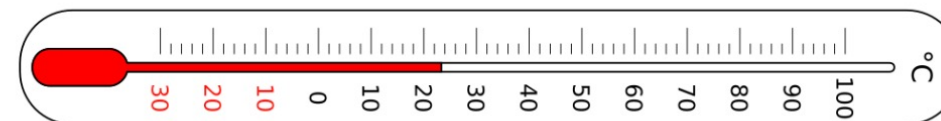
Vojtech Halenka, 2024



# Position encoding



| Features | $X_1$ | $X_2$ | $X_3$ | $X_1$ | $X_2$ | $X_3$ | $X_1$ | $X_2$ | $X_3$ | $X_1$ | $X_2$ | $X_3$ |
|----------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
| Encoding | F     | F     | F     | T     | F     | F     | T     | T     | F     | T     | T     | T     |
| Value    | 0     |       |       | 1     |       |       | 2     |       |       | 3     |       |       |



# Position encoding

**0: U L**

```

00*000*000
*000*0*000
**00*00000
00000000**
00*000****
000000***1
000**0*111
0*00**111*
*****111*0
00**11**00
    
```

---

$0 < X \leq 6$   
 $5 < Y \leq 8$

**0: U R**

```

0*****
0*****0**
*****00
*11111**0
*111*11**0
*1*1*11***
111**11**0
1****111**
11***111**
*****111**
    
```

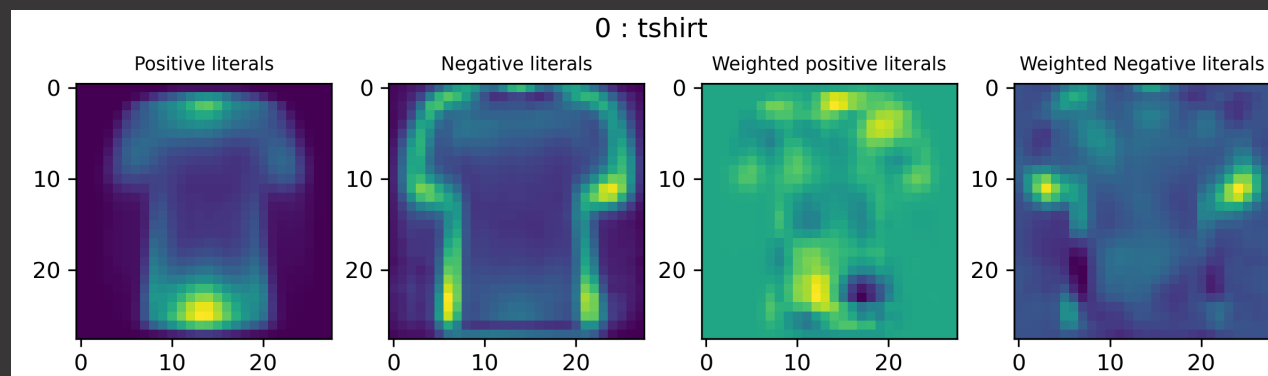
---

$8 < X \leq 17$   
 $0 < Y \leq 6$



# Performance

| Model                               | 2D N-XOR    | MNIST       | K-MNIST      | F-MNIST      |
|-------------------------------------|-------------|-------------|--------------|--------------|
| 4-Nearest Neighbour [5, 41]         | 61.62       | 97.14       | 91.56        | 85.40        |
| SVM [5]                             | 94.63       | 98.57       | 92.82        | 89.7         |
| Random Forest [22]                  | 70.73       | 97.3        | -            | 81.6         |
| Gradient Boosting Classifier [41]   | 87.15       | 96.9        | -            | 88.0         |
| Simple CNN [5, 22]                  | 91.05       | 99.06       | 95.12        | 90.7         |
| BinaryConnect [6]                   | -           | 98.99       | -            | -            |
| FPGA-accelerated BNN [20]           | -           | 98.70       | -            | -            |
| Logistic Circuit (binary) [22]      | -           | 97.4        | -            | 87.6         |
| Logistic Circuit (real-valued) [22] | -           | 99.4        | -            | 91.3         |
| PreActResNet-18 [5]                 | -           | 99.56       | 97.82        | 92.00        |
| ResNet18 + VGG Ensemble [5]         | -           | 99.60       | 98.90        | -            |
| TM                                  | 99.12       | 98.57       | 92.03        | 90.09        |
| CTM (Mean)                          | 99.99 ± 0.0 | 99.33 ± 0.0 | 96.08 ± 0.01 | 91.18 ± 0.01 |
| CTM (95 %ile)                       | 100.0       | 99.38       | 96.25        | 91.39        |
| CTM (Peak)                          | 100.0       | 99.40       | 96.31        | 91.50        |

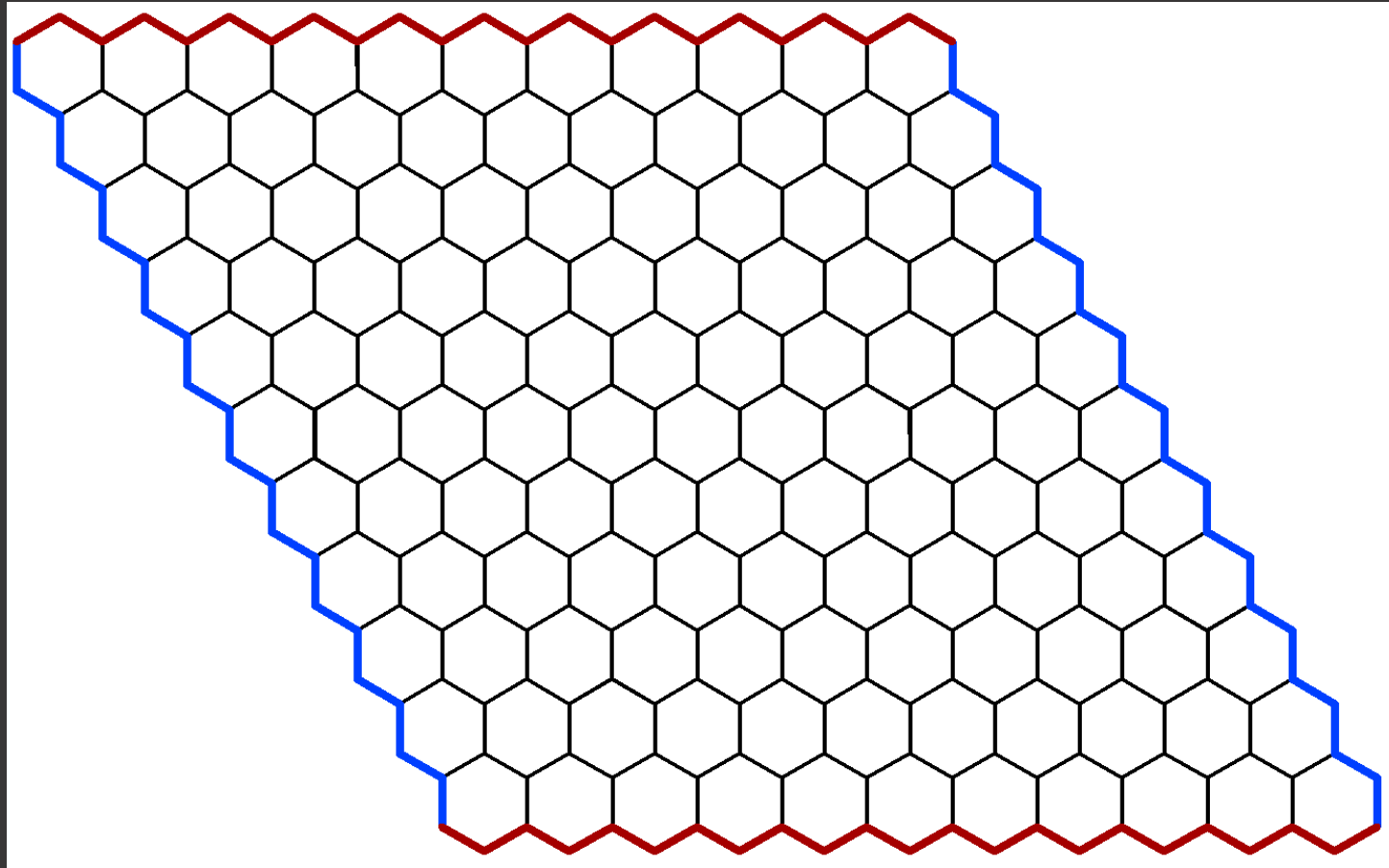


Mayur Shende, 2024



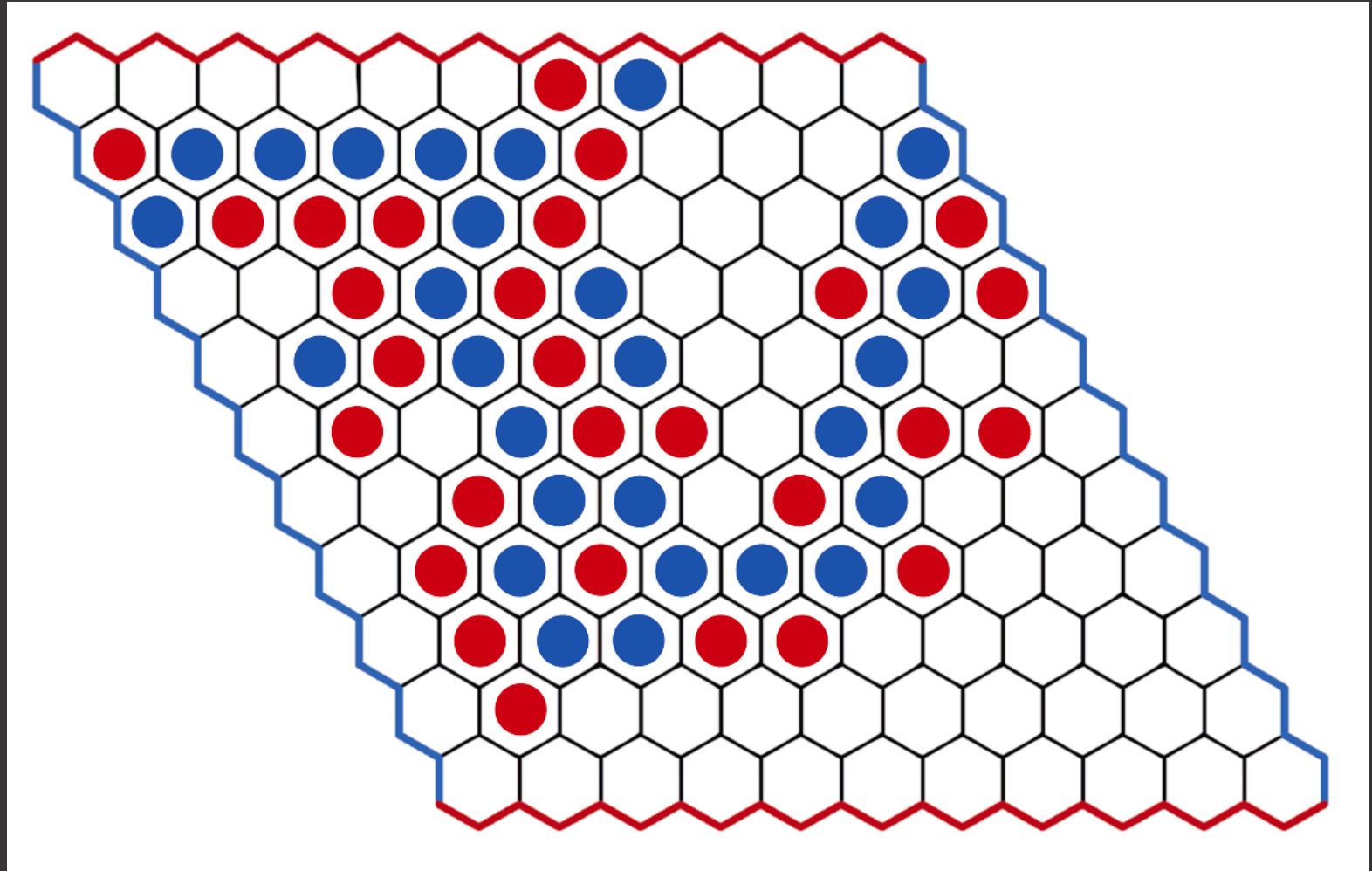
# Board games

# Hex

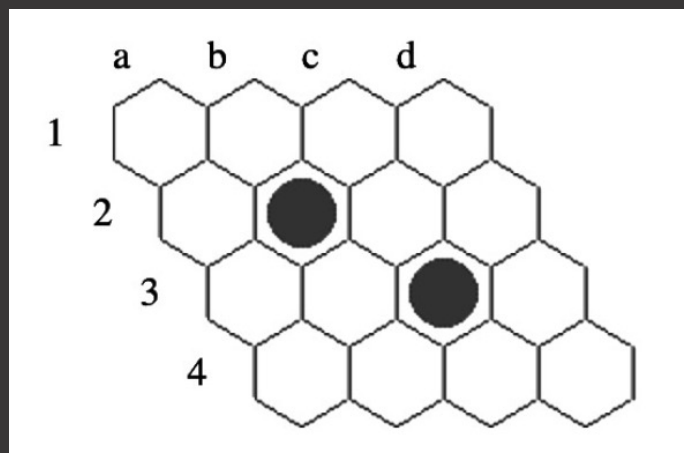


# Hex

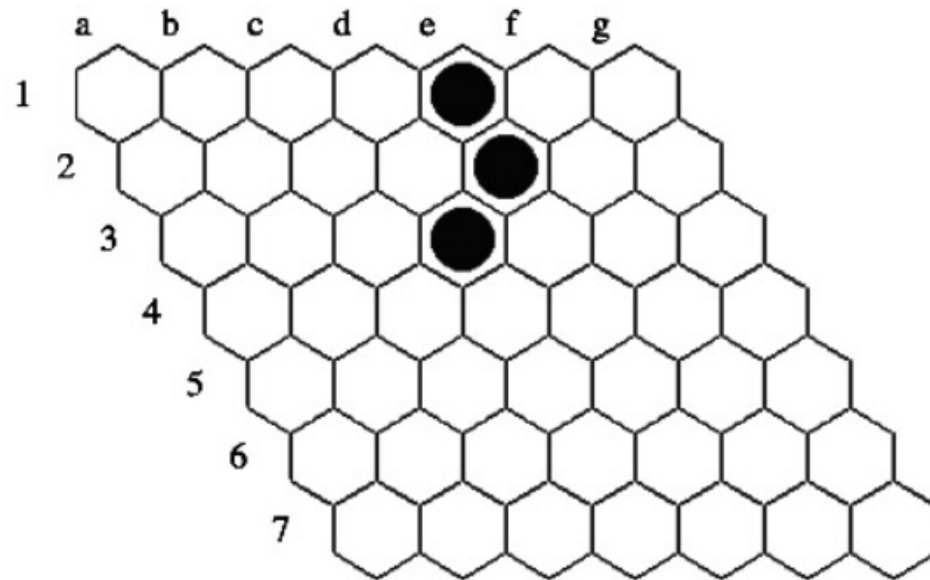
- «Hex can never be a draw.
- Either red or blue will make a chain.
- If the red stones form a chain between the red sides, the blue stones can no longer form a chain between the blue sides, and vice versa.
- Thus, trying to make a chain is the same as preventing a chain of the enemy.
- Therefore, the best defence equals the best attack.
- As soon as you have grasped this aspect of the game, you will see the great beauty of the game and you will be a better player.»



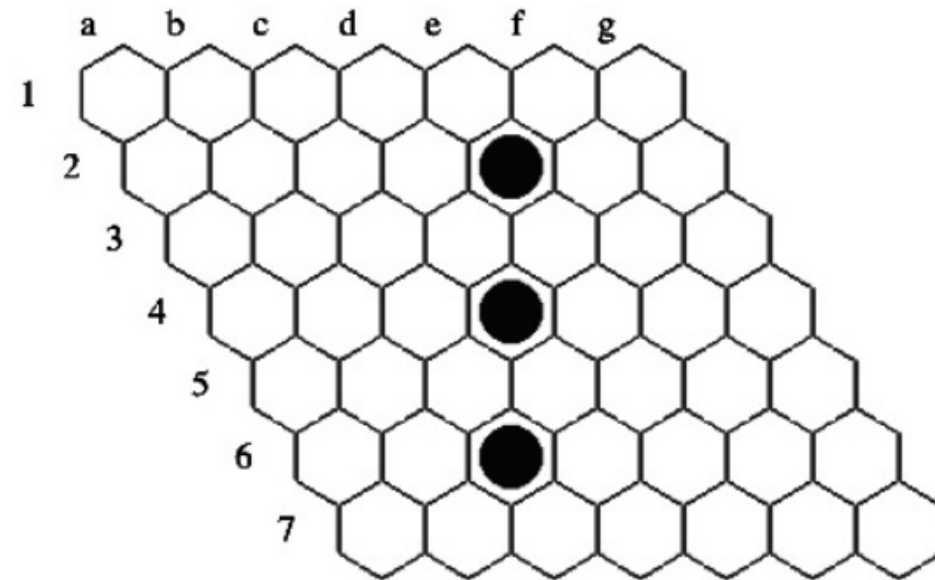
# Bridge pattern



# Bridge pattern



*Expanding by adjacent moves*



*Expanding by two-chains*



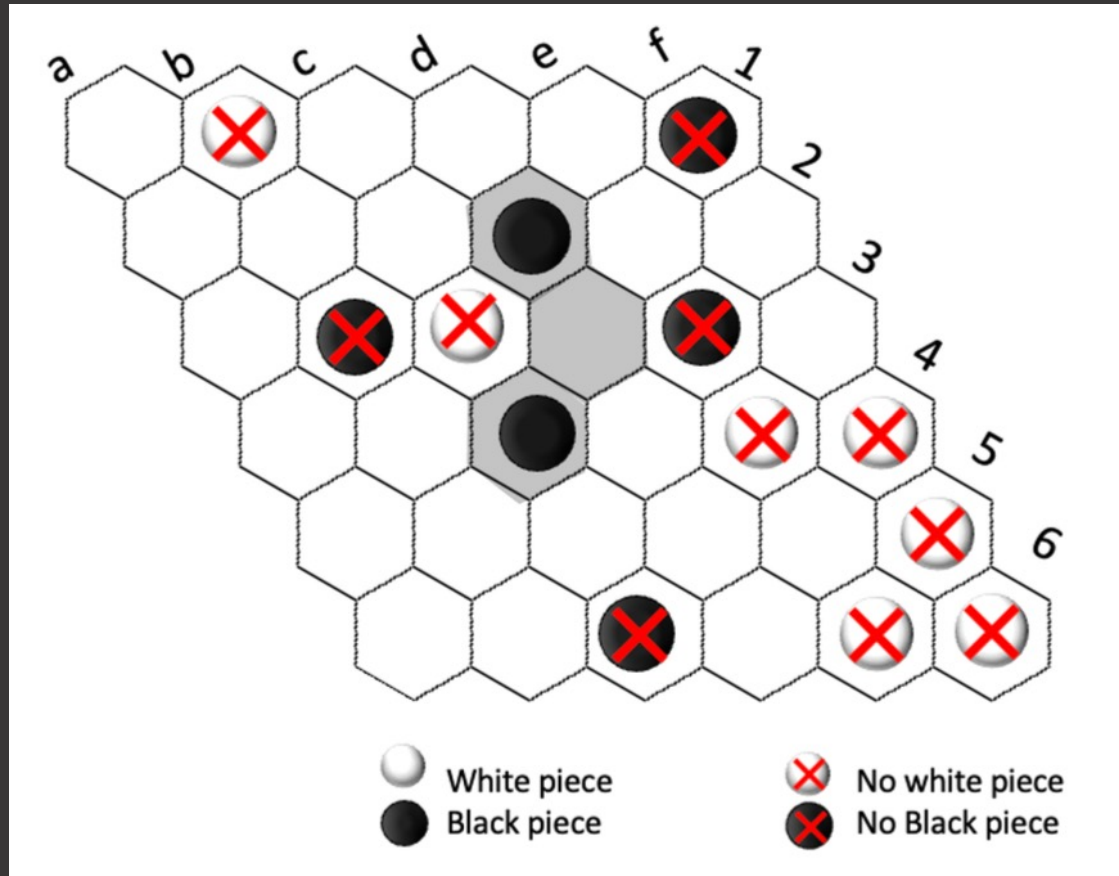
# Strategy

«Thinking Hex is much like thinking Go. Though the aim is to make a chain, it is best to think in areas. Especially at the start of the game, you don't use stones to make chains, but rather occupy areas on the board, thus taking strategic positions. Most of the time, it's better to put stones a bit away from the enemy stones, rather than directly besides them.

The best strategy for making a connection can be formulated as (1) determining your strongest chain and (2) strengthening the weakest link in that chain. The best defending strategy can be formulated as (1) determining the strongest chain of the enemy and (2) weakening the weakest link in that chain. As noted above, the best defence equals the best attack, thus both ways of thinking should result in the same best move.»

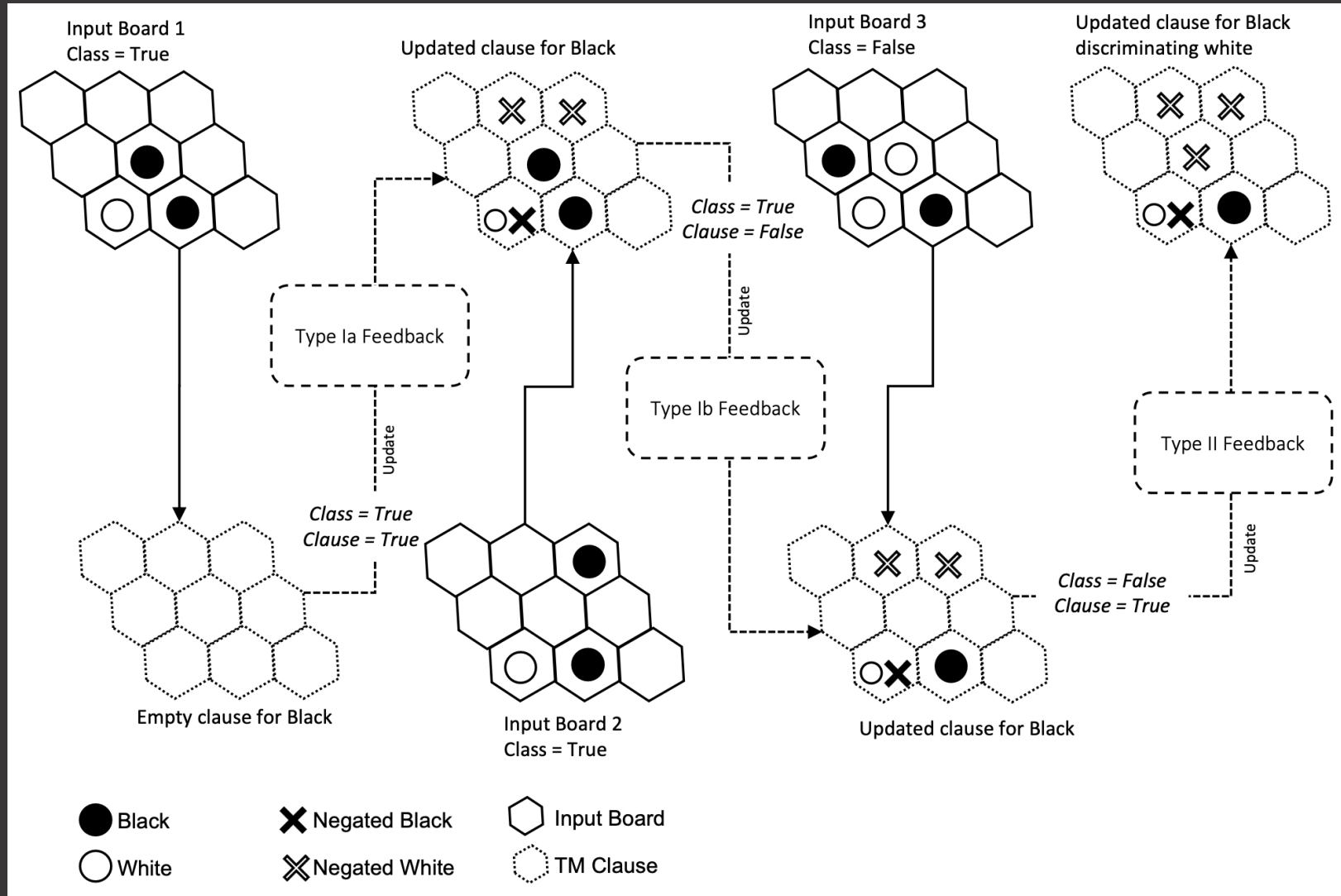
<http://www.lutanho.net/play/hex.html>

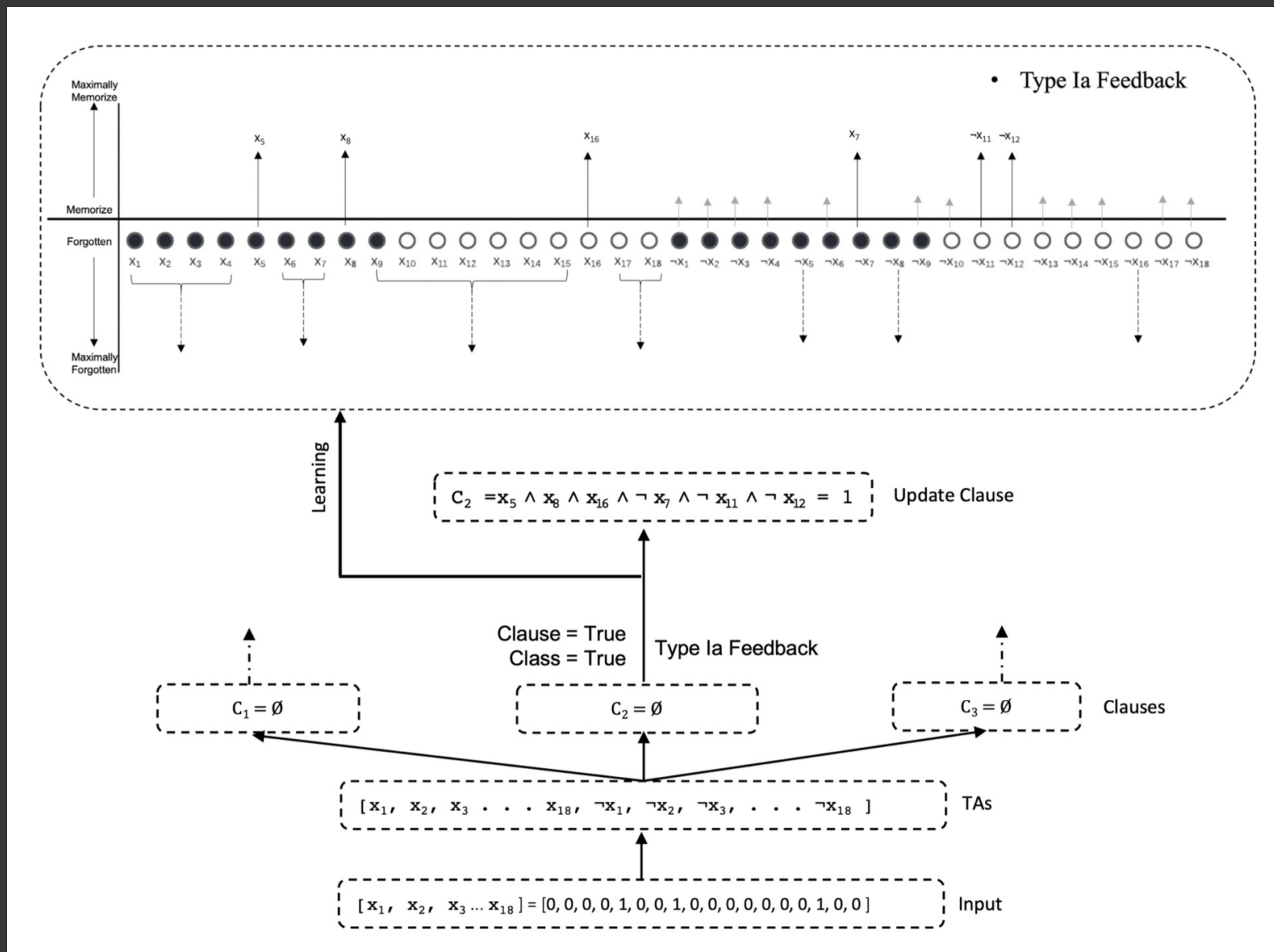
# Example clause for winning black position



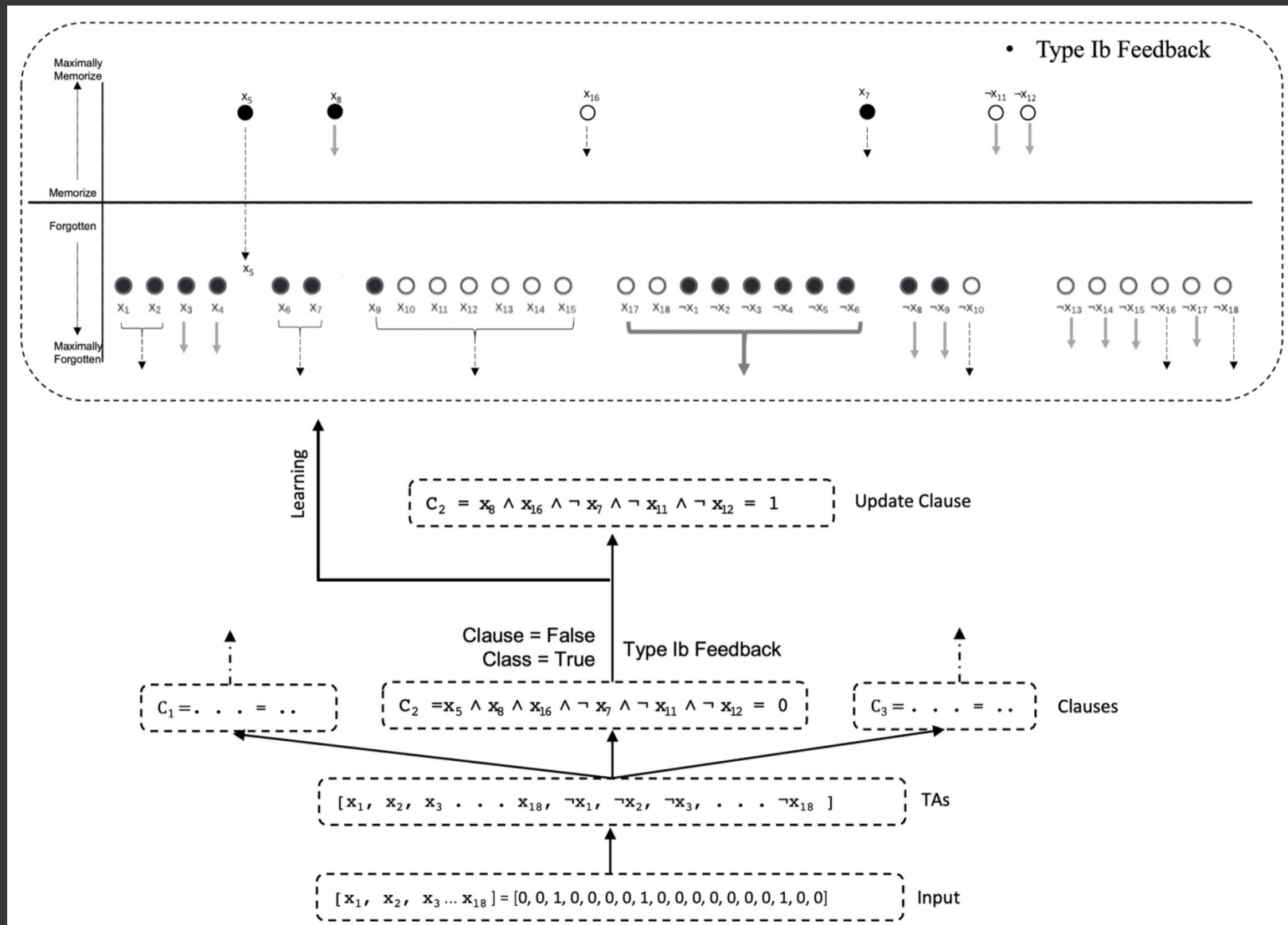
Giri, C., Granmo, O., Hoof, H.V., & Blakely, C.D. (2022). Logic-based AI for Interpretable Board Game Winner Prediction with Tsetlin Machine. *2022 International Joint Conference on Neural Networks (IJCNN)*, 1-9.

# Learning to recognize a winning position



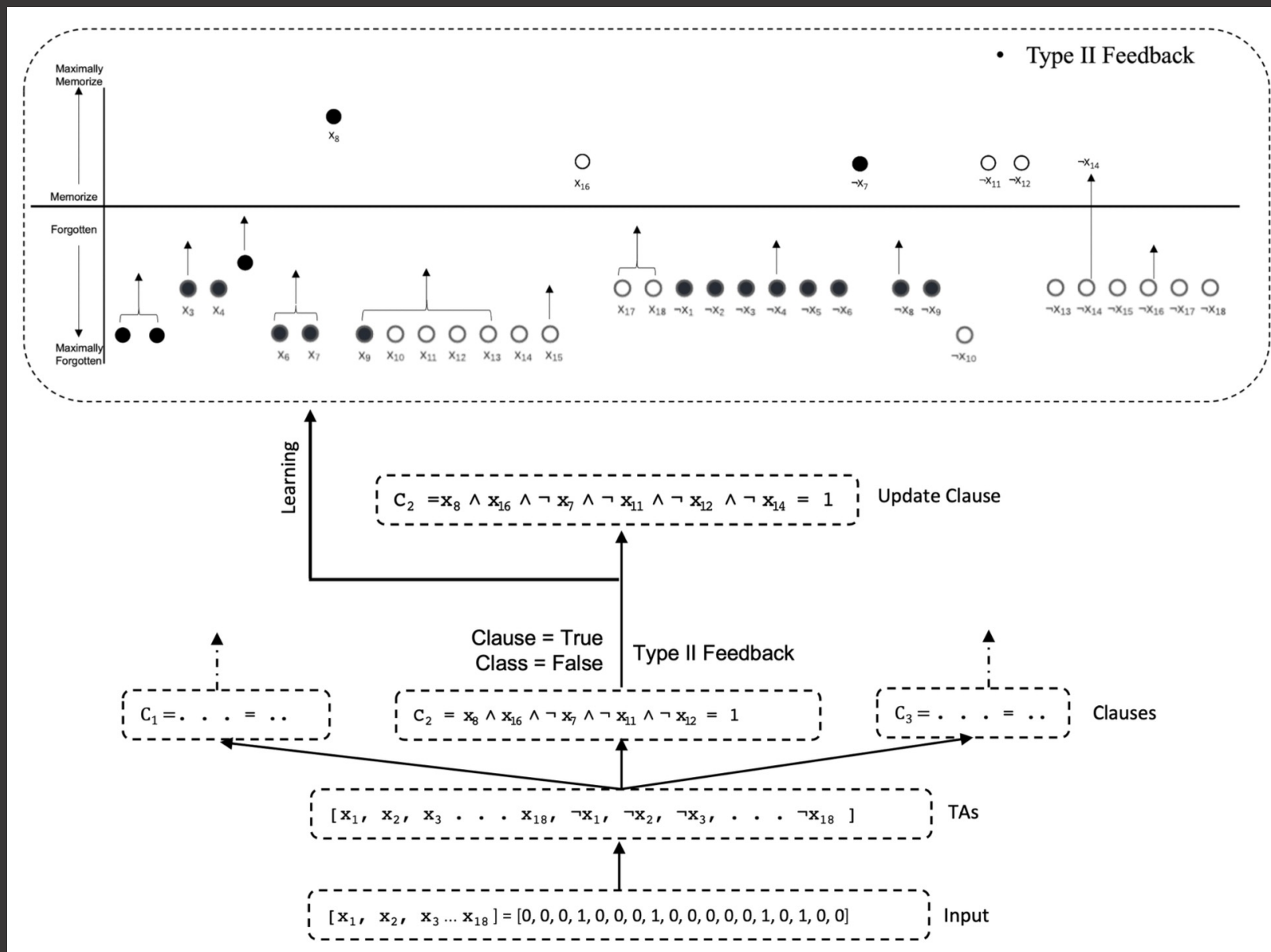


Recognize feedback

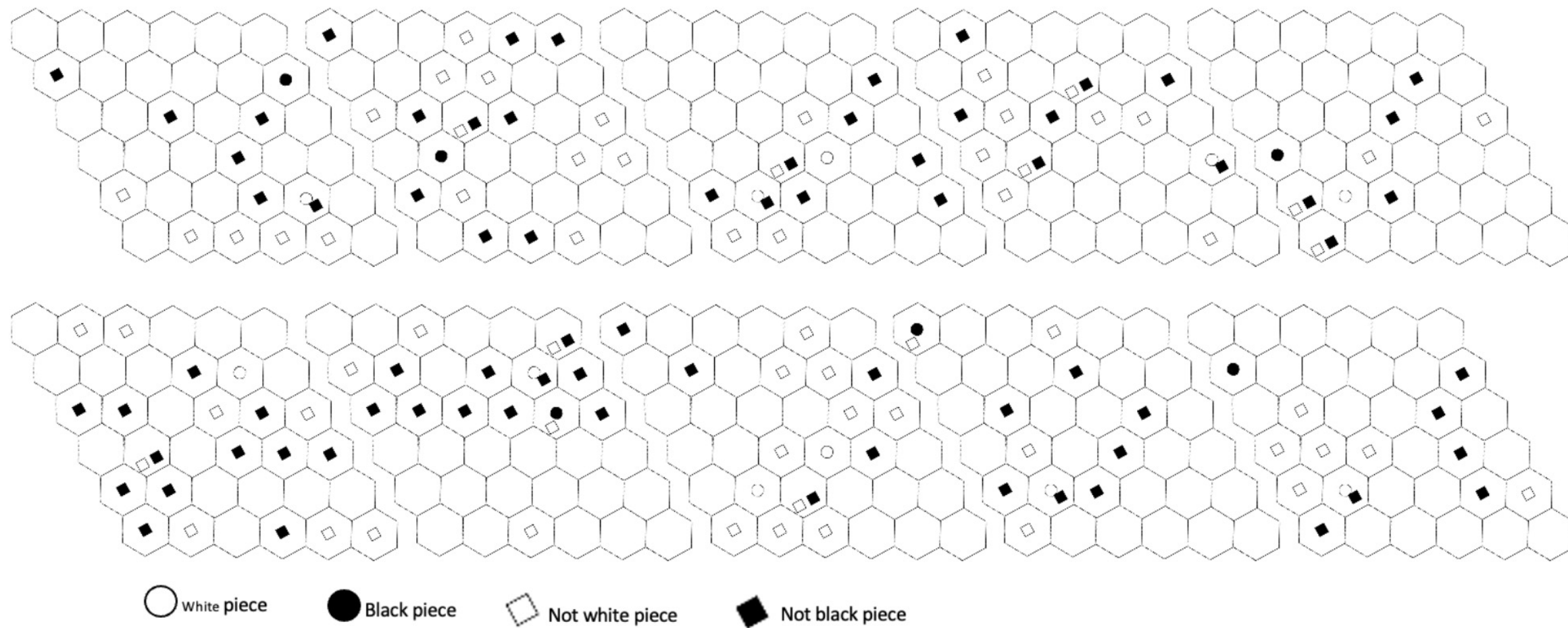


Erase feedback

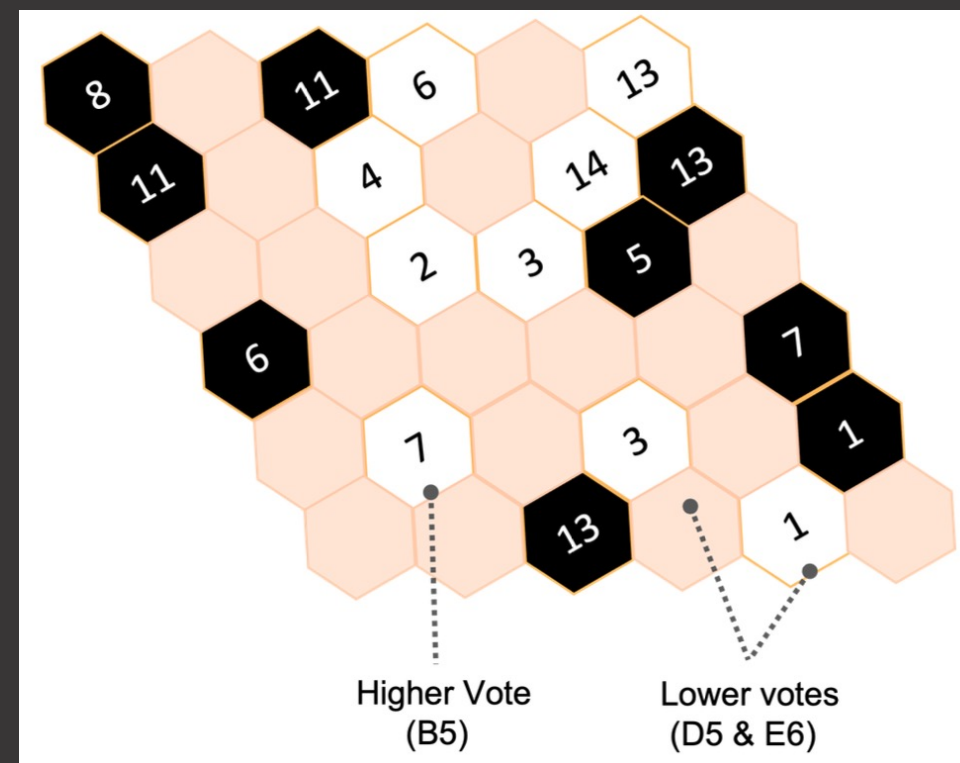
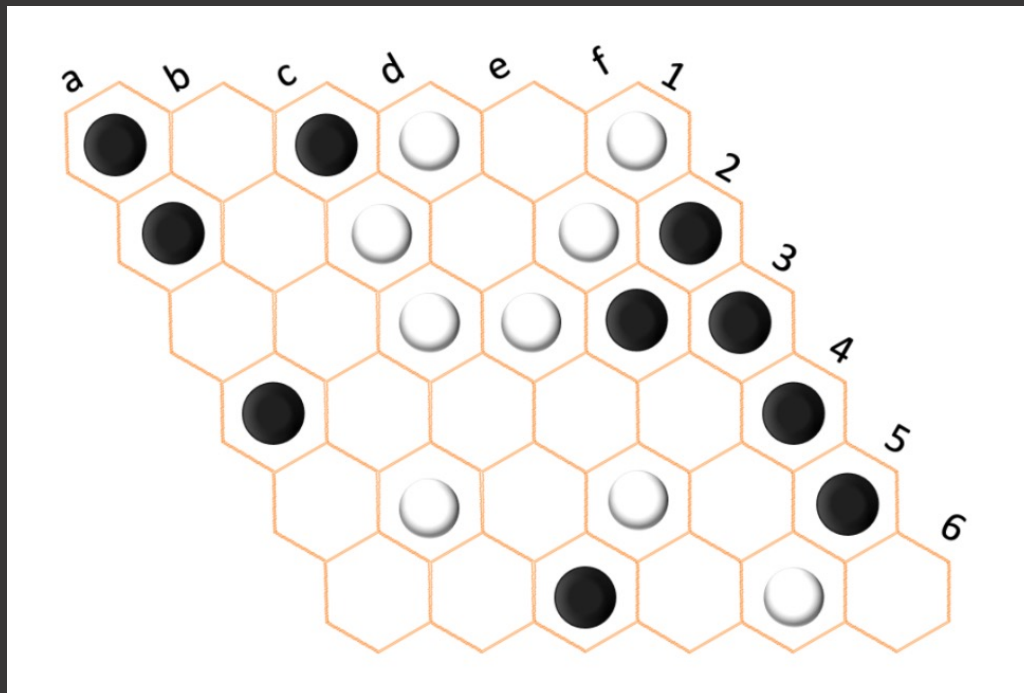




Reject feedback

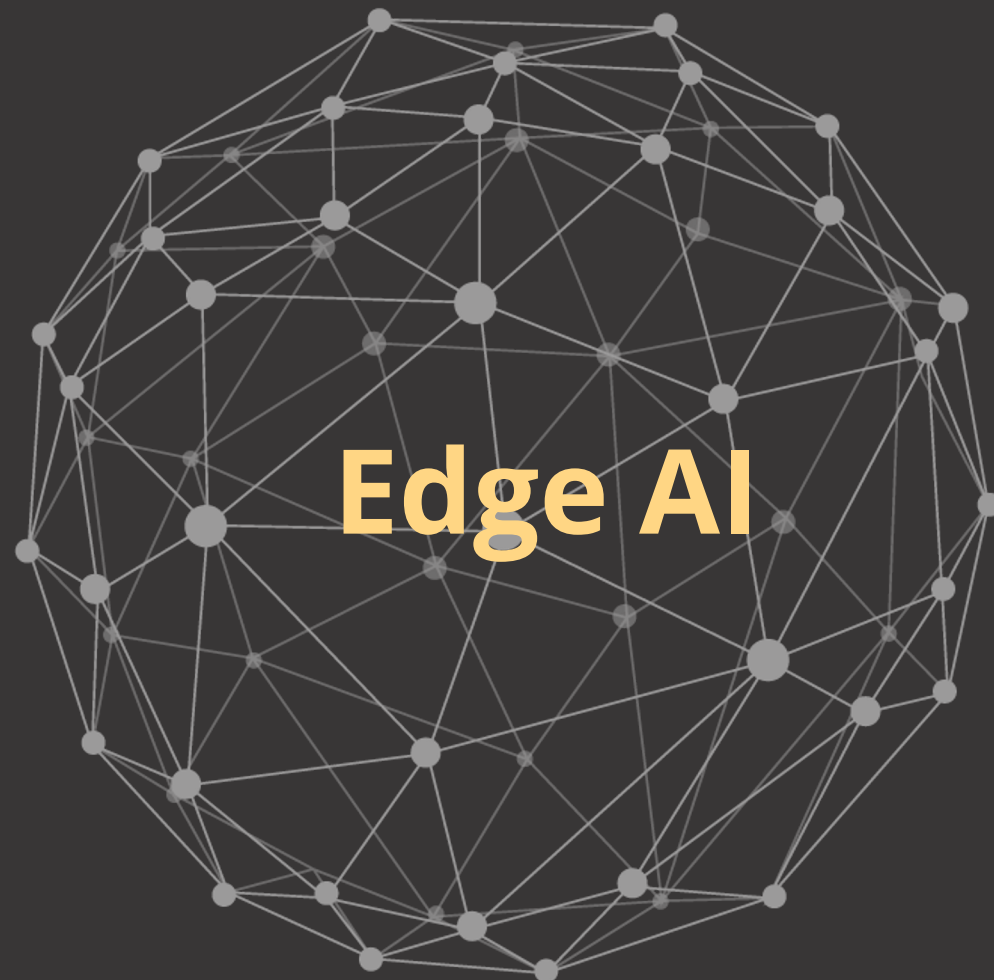


Ten most impactful negative polarity clauses

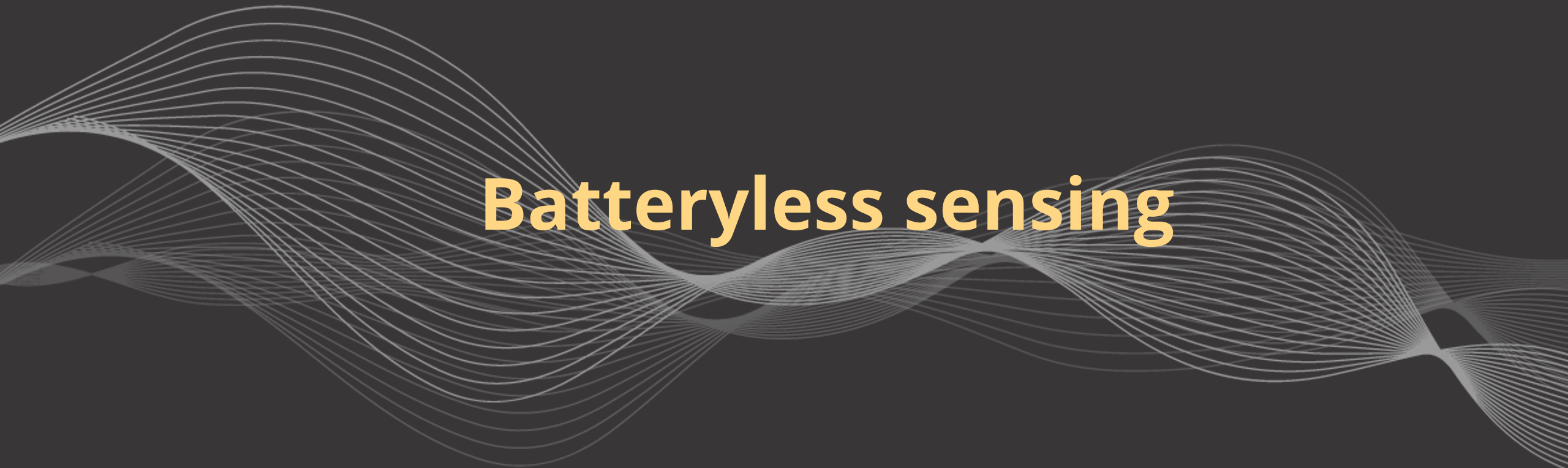


Local interpretation

| METHOD              | HYPERPARAMETER  | TRAINING ACCURACY | TESTING ACCURACY |
|---------------------|-----------------|-------------------|------------------|
| NAÏVE BAYES         | DEFAULT         | 72.02             | 72.02            |
| DECISION TREE       | MAX DEPTH=100   | 100               | 87.1             |
| K-NEAREST NEIGHBOUR | K=1             | 100               | 84               |
| XGBOOST             |                 | 84.88             | 88.90            |
| INTERPRETML(EBC)    | DEFAULT         | 74.56             | 74.33            |
| NEURAL NETWORK      |                 | 97.6              | 90.7             |
|                     | CLAUSES = 10000 |                   |                  |
|                     | T = 8000        |                   |                  |
|                     | S = 100         |                   |                  |
| TSETLIN MACHINE     | MAX WEIGHT=255  | 95.6              | 92.1             |



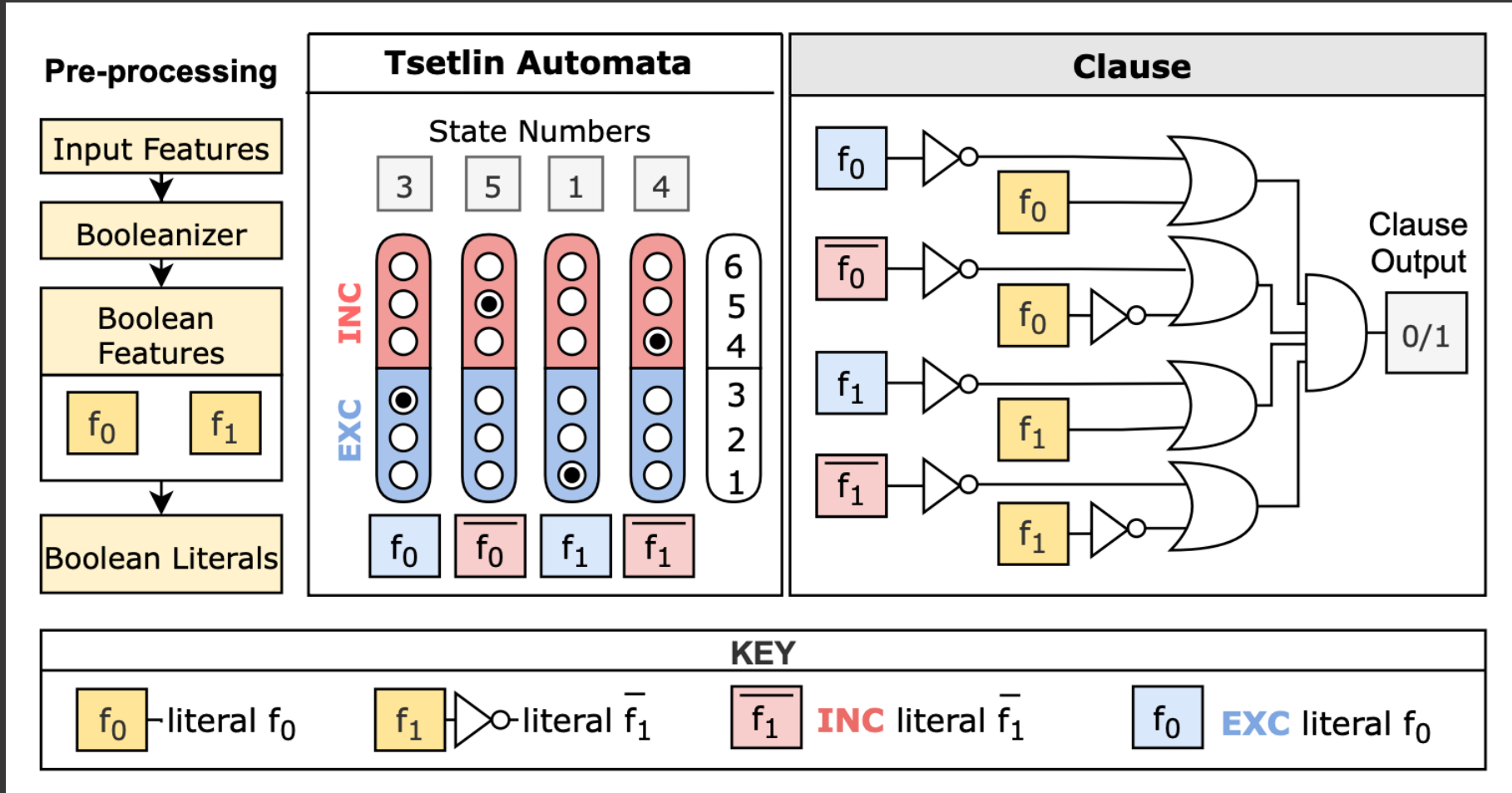
**Edge AI**



# Batteryless sensing



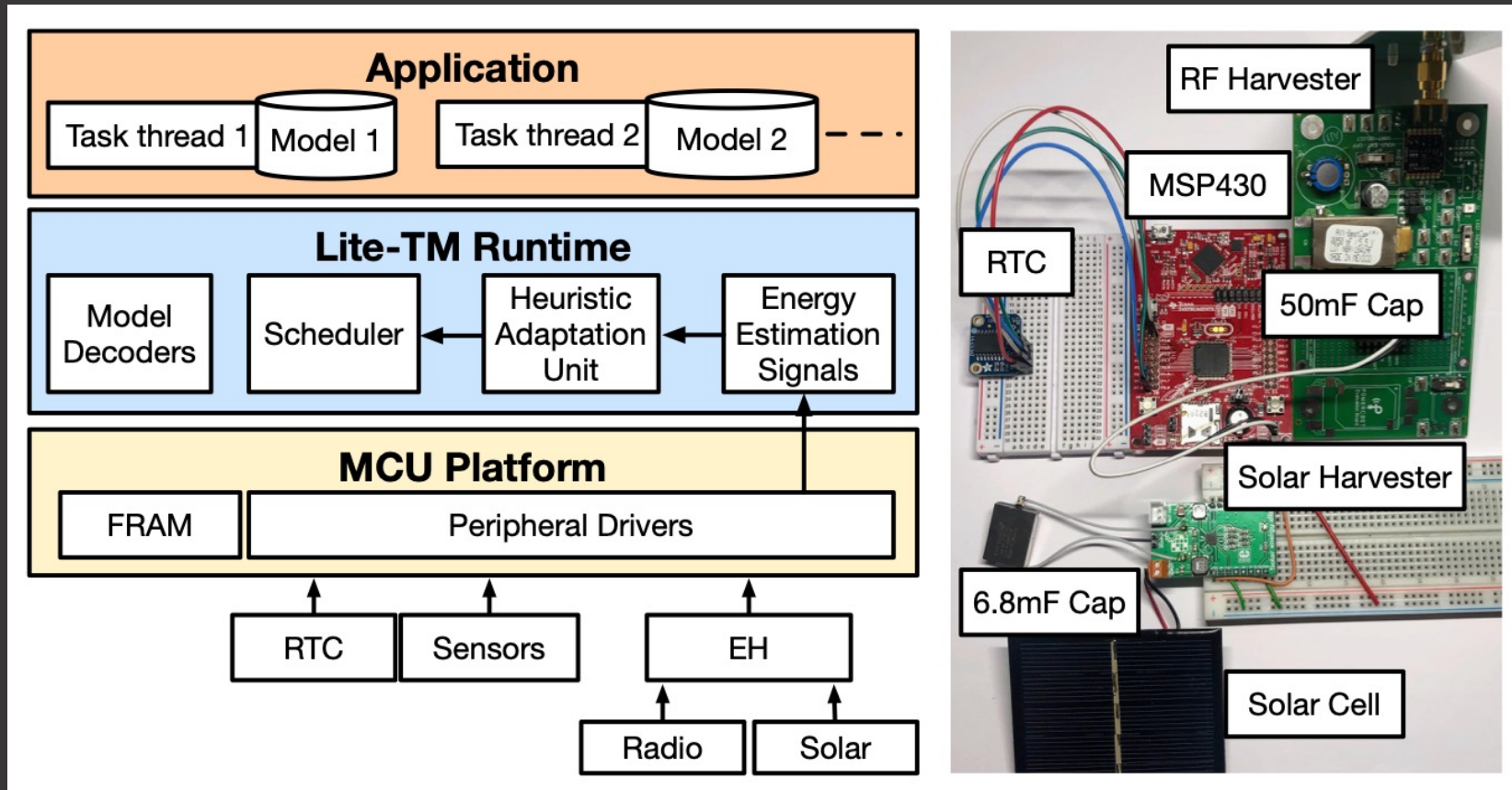
# Hardware-near learning and inference



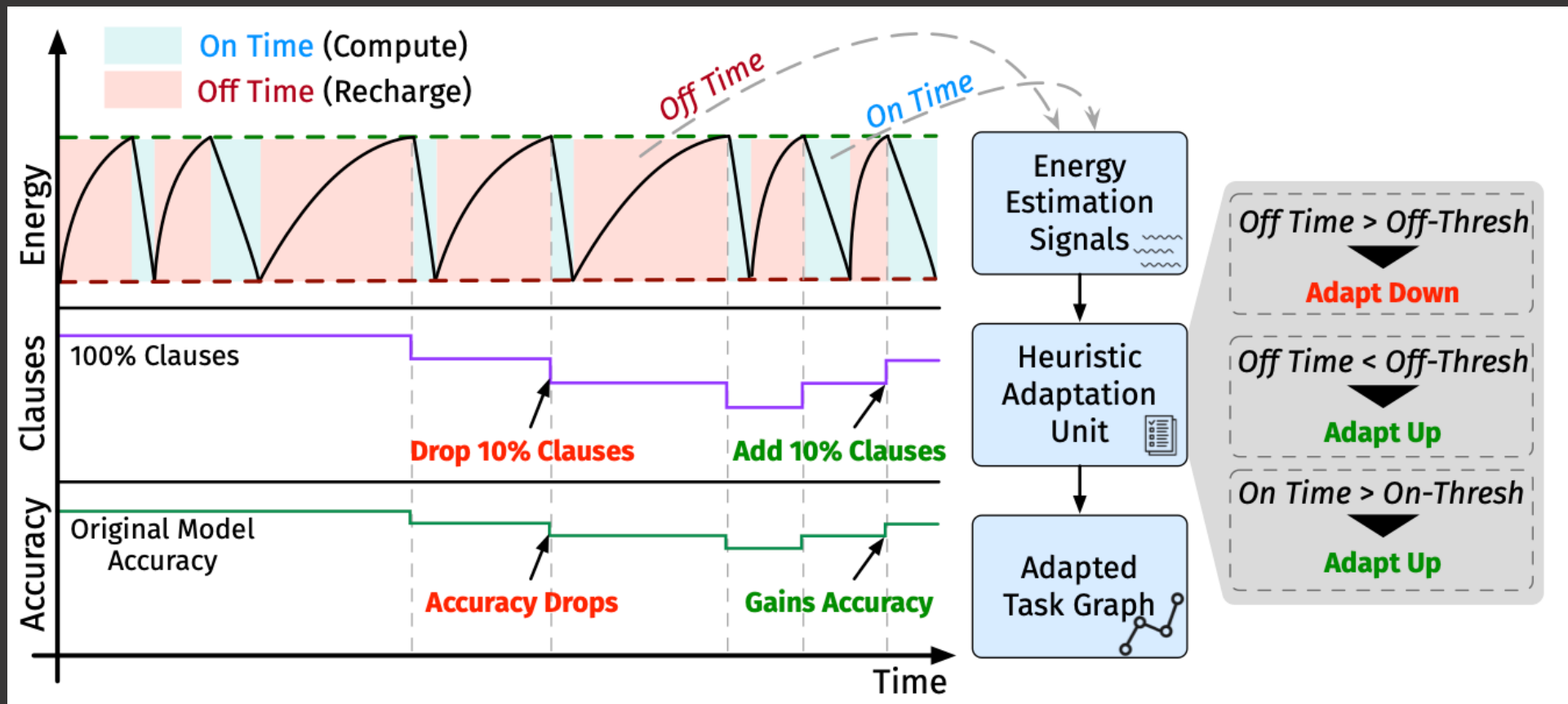
Abu Bakar, Tousif Rahman, Rishad Shafik, Fahim Kawsar, and Alessandro Montanari. Adaptive Intelligence for Batteryless Sensors Using Software-Accelerated Tsetlin Machines. In Proceedings of SenSys 2022. ACM, 2022.

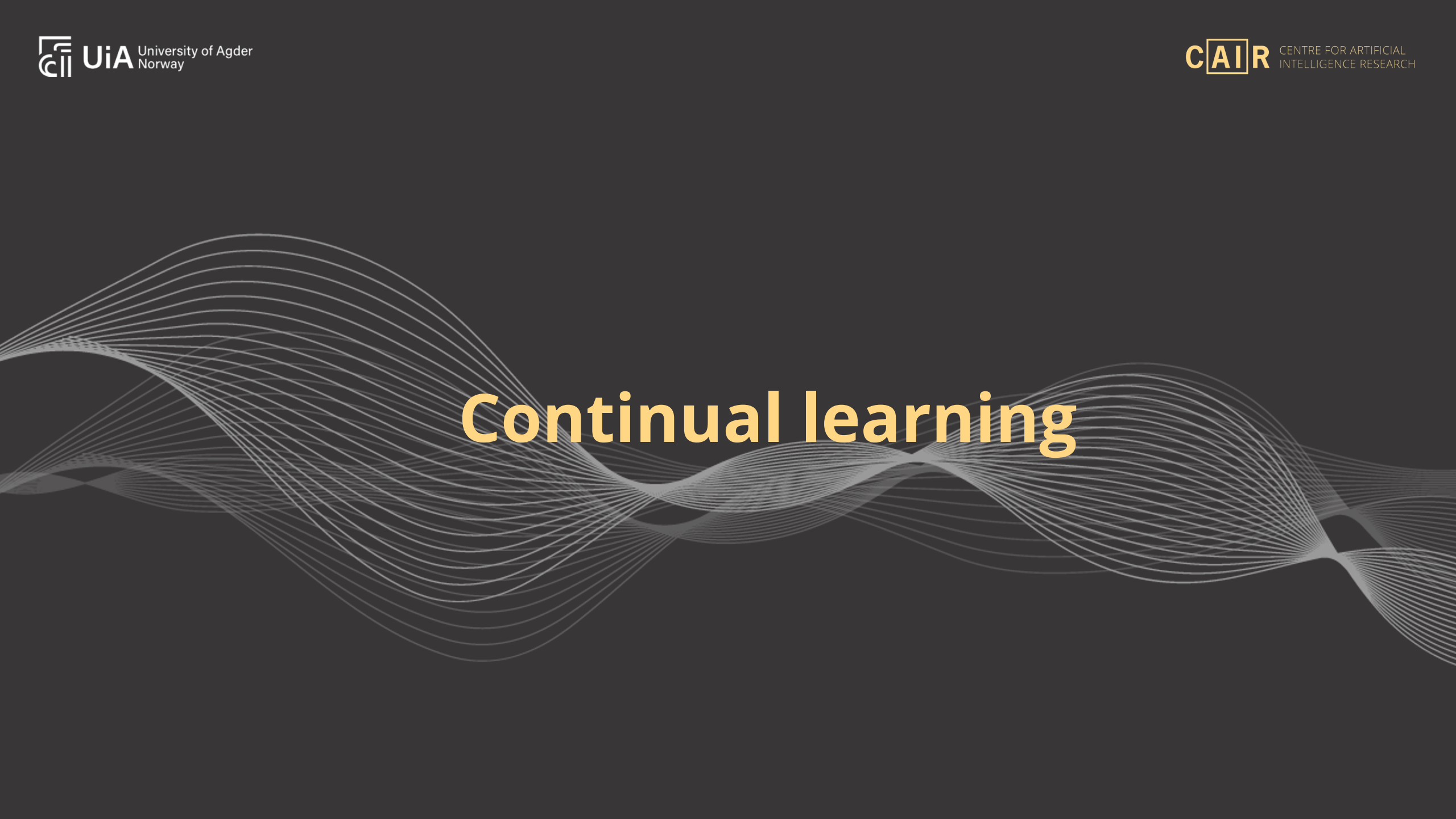


# Batteryless sensing

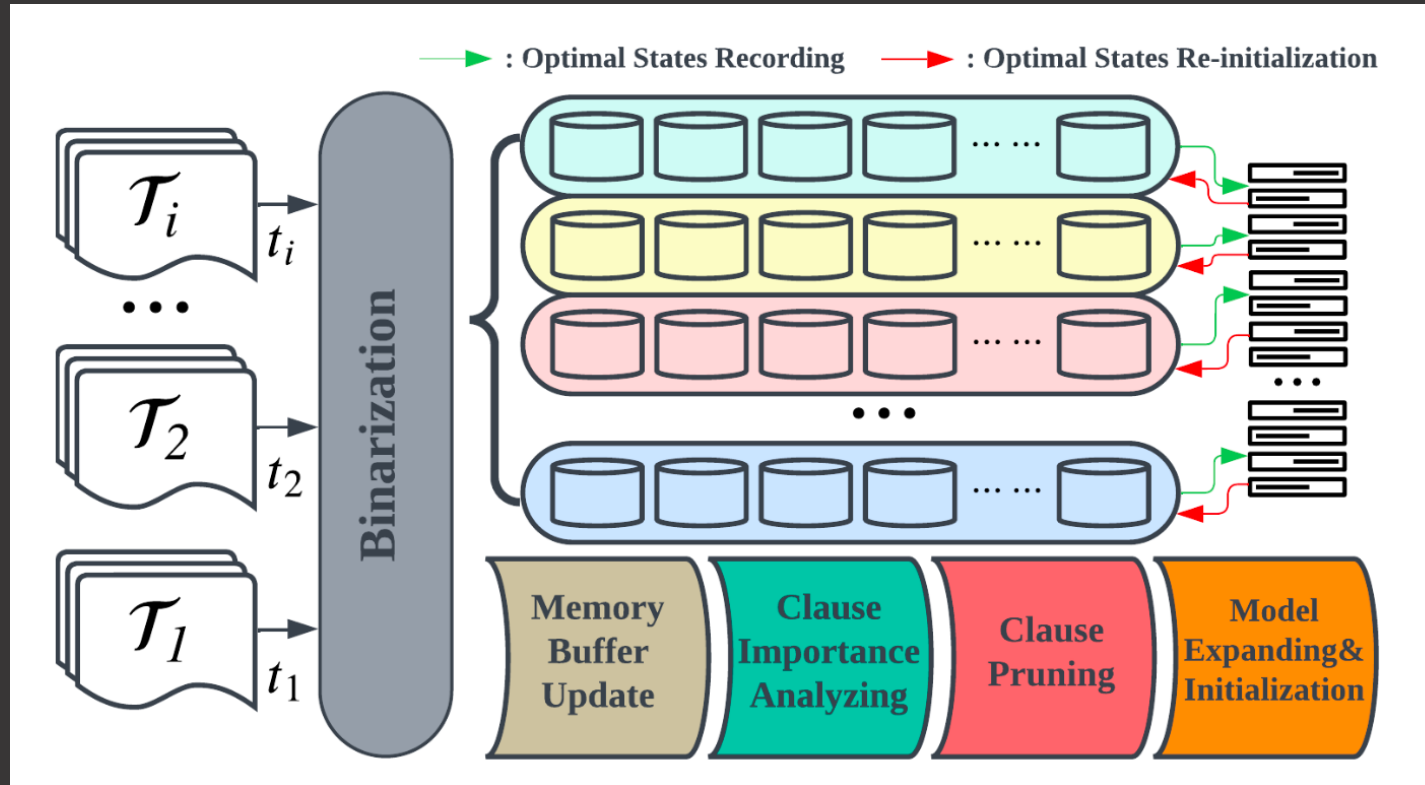


# Batteryless sensing





# Continual learning



Chong Tang, Neelam Singh, Jagmohan Chauhan. AdaTM: Logic Inspired Adaptive Tsetlin Machines for Efficient and Effective Continual Learning on the Edge. EWSN '24, December 10–13, 2024, Abu Dhabi, United Arab Emirates

«Continuous learning, crucial for applications with sequentially arriving data, enables dynamically acquiring new skills without forgetting previous knowledge.»

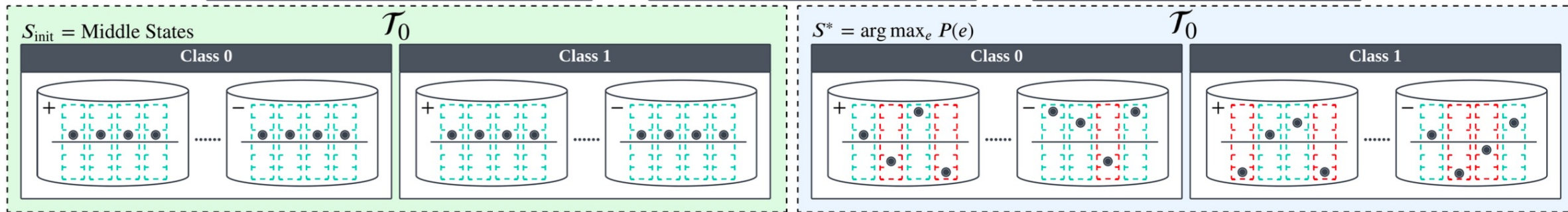


Two clause teams are assigned to two classes in task 0. TAs of each are initialized to be middle states.

One-vs-all  
Training

$$y_k(i) = \begin{cases} 1 & \text{if } y(i) = k \\ 0 & \text{otherwise} \end{cases}$$

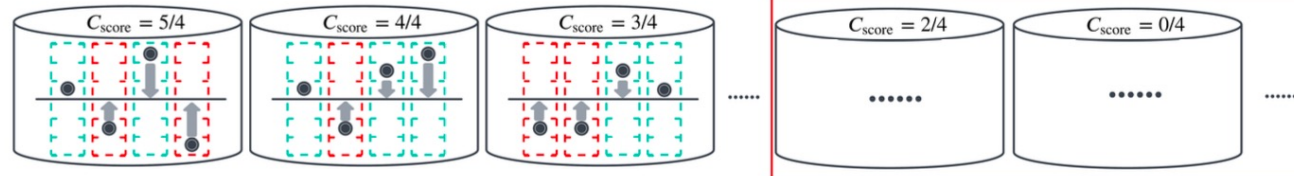
Training TAs and performance control.



Clause Confidence-based Pruning:

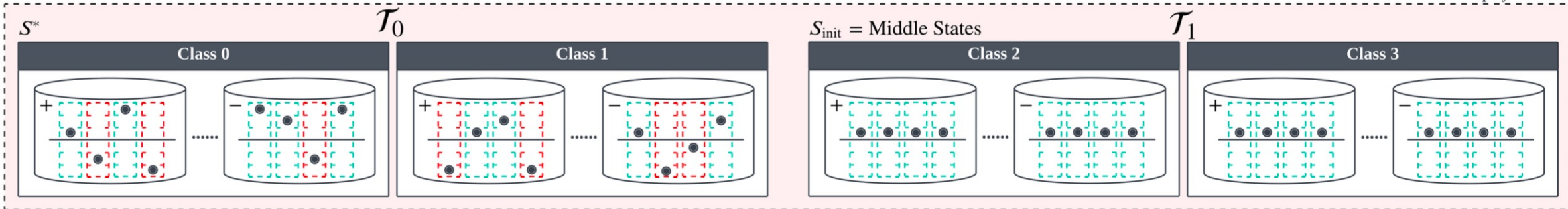
$$C_{\text{score}} = \frac{1}{n} \sum_{i=1}^n |\tau_i - \theta|$$

$$\mathcal{P} = \text{Prune}_N(\mathcal{C}, C_{\text{score}})$$



$X \subseteq \mathcal{T}_0, |X| = N$   
for each class in  $\mathcal{T}_0$

Memory Update  
and Repaly



| Strategies  | Models   | MNIST       |            | FashionMNIST |            | AudioMNIST  |            | TESS        |            | PAMAP2      |            |
|-------------|----------|-------------|------------|--------------|------------|-------------|------------|-------------|------------|-------------|------------|
|             |          | $ACC_{avg}$ | $FM_{avg}$ | $ACC_{avg}$  | $FM_{avg}$ | $ACC_{avg}$ | $FM_{avg}$ | $ACC_{avg}$ | $FM_{avg}$ | $ACC_{avg}$ | $FM_{avg}$ |
| EWC         | MLP      | 19.43%      | 99.82%     | 19.93%       | 100.00%    | 20.10%      | 99.08%     | 25.00%      | 94.44%     | 13.07%      | 88.25%     |
|             | LeNet    | 22.77%      | 83.51%     | 21.89%       | 96.47%     | 21.33%      | 86.65%     | 55.31%      | 75.73%     | -           | -          |
|             | ResNet   | 19.72%      | 100.00%    | 19.96%       | 100.00%    | 19.88%      | 100.00%    | 48.85%      | 100.00%    | -           | -          |
|             | B-MLP    | 21.94%      | 84.81%     | 23.22%       | 93.29%     | 26.30%      | 78.41%     | 52.17%      | 73.84%     | 14.70%      | 81.64%     |
|             | B-LeNet  | 19.79%      | 95.59%     | 19.30%       | 99.87%     | 19.57%      | 99.22%     | 48.83%      | 92.46%     | -           | -          |
|             | B-ResNet | 16.92%      | 100.00%    | 18.46%       | 100.00%    | 24.93%      | 91.44%     | 38.67%      | 95.91%     | -           | -          |
| GDumb       | MLP      | 85.97%      | 2.72%      | 72.12%       | 7.55%      | 77.07%      | 7.12%      | 57.26%      | 19.66%     | 43.42%      | 14.47%     |
|             | LeNet    | 85.52%      | 2.32%      | 71.30%       | 7.16%      | 87.42%      | 2.98%      | 56.43%      | 21.01%     | -           | -          |
|             | ResNet   | 93.92%      | 1.19%      | 78.79%       | 5.50%      | 96.25%      | 0.71%      | 26.56%      | 50.00%     | -           | -          |
|             | B-MLP    | 73.73%      | 4.18%      | 65.50%       | 9.10%      | 55.69%      | 10.25%     | 38.38%      | 25.69%     | 23.60%      | 16.97%     |
|             | B-LeNet  | 78.32%      | 5.01%      | 51.58%       | 15.25%     | 61.32%      | 8.66%      | 54.56%      | 25.53%     | -           | -          |
|             | B-ResNet | 14.51%      | -0.46%     | 25.99%       | 26.28%     | 66.45%      | 6.48%      | 43.36%      | 22.31%     | -           | -          |
| DGR         | MLP      | 53.61%      | 26.71%     | 47.46%       | 43.95%     | 24.95%      | 84.40%     | 35.48%      | 100.00%    | 23.18%      | 67.43%     |
|             | LeNet    | 36.99%      | 32.94%     | 20.09%       | 96.94%     | 29.10%      | 72.86%     | 34.65%      | 100.00%    | -           | -          |
|             | ResNet   | 31.00%      | 42.17%     | 19.81%       | 100.00%    | 19.78%      | 100.00%    | 24.90%      | 100.00%    | -           | -          |
|             | B-MLP    | 54.71%      | 12.38%     | 36.21%       | 51.87%     | 32.65%      | 66.08%     | 20.12%      | 100.00%    | 19.26%      | 51.07%     |
|             | B-LeNet  | 50.99%      | 24.49%     | 44.74%       | 31.71%     | 34.05%      | 57.00%     | 30.91%      | 100.00%    | -           | -          |
|             | B-ResNet | 23.15%      | 57.28%     | 10.00%       | 85.16%     | 23.88%      | 85.50%     | 28.42%      | 100.00%    | -           | -          |
| ER          | MLP      | 90.57%      | 3.05%      | 80.55%       | 8.38%      | 71.14%      | 10.80%     | 50.21%      | 48.25%     | 38.96%      | 18.00%     |
|             | LeNet    | 96.24%      | 1.07%      | 81.80%       | 8.05%      | 93.39%      | 1.62%      | 82.57%      | 20.01%     | -           | -          |
|             | ResNet   | 95.92%      | 1.62%      | 79.03%       | 11.96%     | 93.55%      | 2.02%      | 37.76%      | 65.85%     | -           | -          |
|             | B-MLP    | 83.98%      | 6.25%      | 73.49%       | 13.40%     | 74.03%      | 6.26%      | 70.33%      | 27.27%     | 27.65%      | 27.34%     |
|             | B-LeNet  | 89.59%      | 2.93%      | 66.65%       | 9.88%      | 74.75%      | 4.83%      | 72.61%      | 21.25%     | -           | -          |
|             | B-ResNet | 83.25%      | 3.14%      | 72.44%       | 11.99%     | 89.37%      | 3.26%      | 54.15%      | 38.53%     | -           | -          |
| iCaRL       | MLP      | 80.88%      | 2.62%      | 70.20%       | 7.44%      | 73.11%      | 5.90%      | 62.04%      | 18.41%     | 49.39%      | 2.94%      |
|             | LeNet    | 76.11%      | 1.91%      | 64.63%       | 7.78%      | 67.03%      | 8.63%      | 65.43%      | 6.16%      | -           | -          |
|             | ResNet   | 83.04%      | 1.44%      | 72.37%       | 7.25%      | 90.97%      | 2.34%      | 82.81%      | -7.50%     | -           | -          |
|             | B-MLP    | 78.84%      | 1.96%      | 66.98%       | 13.06%     | 68.71%      | 8.23%      | 79.51%      | 4.65%      | 30.82%      | 9.41%      |
|             | B-LeNet  | 46.96%      | 13.94%     | 44.94%       | 24.06%     | 41.04%      | 15.66%     | 53.92%      | 31.08%     | -           | -          |
|             | B-ResNet | 38.26%      | 10.23%     | 40.57%       | 15.69%     | -           | -          | -           | -          | -           | -          |
| AdaTM       | TM       | 93.88%      | 2.59%      | 80.55%       | 12.54%     | 90.45%      | 3.56%      | 93.36%      | 7.43%      | 70.82%      | 15.64%     |
|             | CTM      | 97.03%      | 1.25%      | 82.67%       | 8.16%      | 89.15%      | 4.08%      | 88.80%      | 10.88%     | -           | -          |
|             | WTM      | 93.87%      | 2.35%      | 80.79%       | 10.08%     | 91.21%      | 2.91%      | 96.26%      | 3.62%      | 74.80%      | -3.00%     |
|             | WCTM     | 97.29%      | 1.04%      | 82.33%       | 9.35%      | 88.92%      | 3.52%      | 91.29%      | 8.53%      | -           | -          |
| AdaTM+Prune | WTM      | 92.54%      | 2.88%      | 80.53%       | 11.28%     | 90.99%      | 2.82%      | 98.55%      | 0.49%      | 73.02%      | -8.66%     |

# Next lecture:

